5 A global 1D array `Num` of integers contains four elements. A program assigns values to these elements as shown:

```
Num[1] ← 1
Num[2] ← 2
Num[3] ← 5
Num[4] ← 3
```

A procedure `Process()` manipulates the values in the array.

The procedure is written in pseudocode:

```
PROCEDURE Process(Start : INTEGER)
   DECLARE CaseVar, Index, Count : INTEGER

   Index ← Start
   Count ← 0

   WHILE Count <= 20
      CaseVar ← Num[Index]
      CASE OF CaseVar
         1 : Num[Index] ← Num[Index] + Index //clause one
             Index ← Index + 1
             Count ← Count + 1

         2 : Num[Index] ← Num[Index] + Index //clause two
             Index ← Index + 2
             Count ← Count + 2

         3 : Num[Index] ← Num[Index] * 2     //clause three
             Index ← Index + 3
             Count ← Count - 1

         4 : Count ← Count + 4               //clause four
         OTHERWISE : Count ← 20
      ENDCASE

      Index ← (Index MOD 4) + 1

   ENDWHILE

ENDPROCEDURE
```

(a) Complete the trace table by dry running the procedure when it is called in the statement:
`CALL Process(1)`

Index	CaseVar	Count	Num[1]	Num[2]	Num[3]	Num[4]

[5]

6 Students are learning about a simple check digit method for data validation. In this method, a single check digit is appended to the end of an original number to give a new number.

The students are studying a method which:

- calculates the sum of all the digits in the original number
- uses integer division to calculate the remainder when the sum is divided by 10
- uses the remainder as the check digit
- appends the check digit to the original number, creating the new number.

For example:

original number	4162
sum of all digits	4 + 1 + 6 + 2 = 13
remainder when the sum is divided by 10 using integer division	3
new number	41623

The original number is always at least three digits in length.

When the new number is input, the check digit is used to validate the new number.

(a) A function `Generate()` is written to take an original number as a parameter and to return the check digit.

Outline a test plan that could be used to fully test function `Generate()`

Assume that the parameter is valid.

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..... [2]

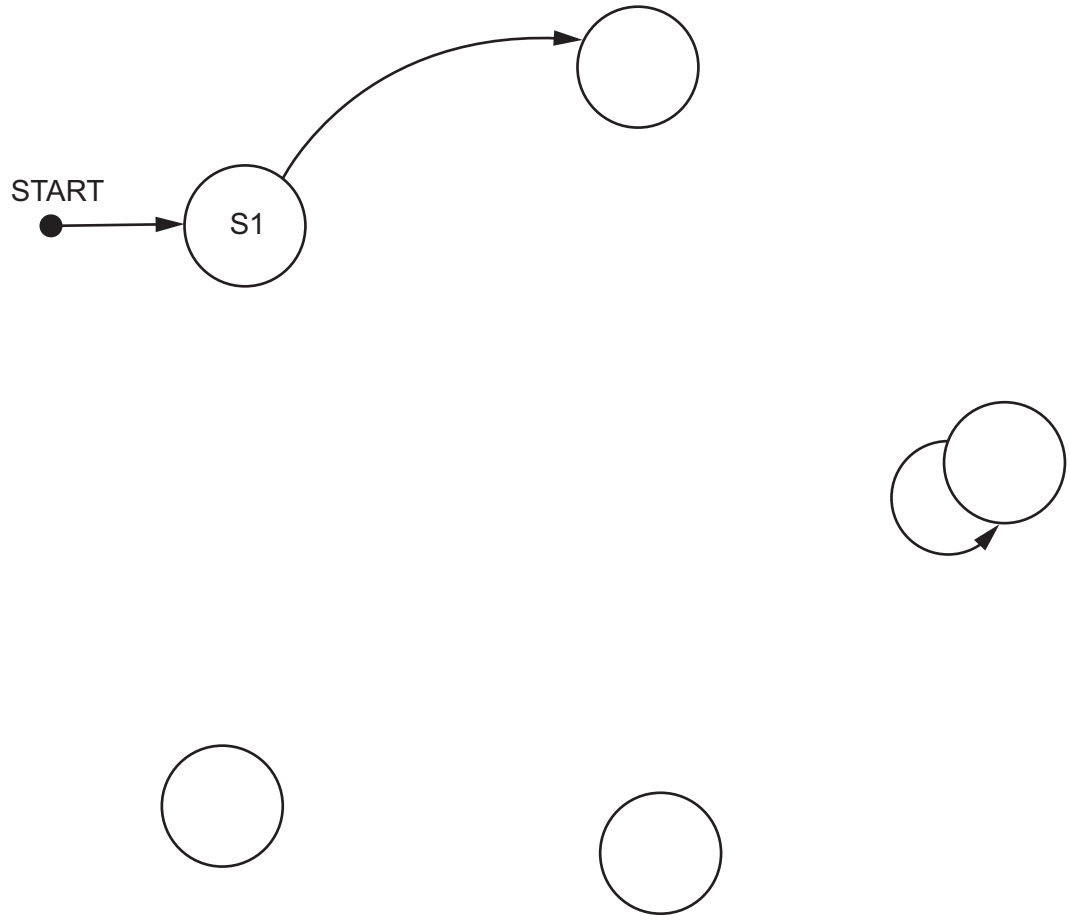
7 There are several different ways to express an algorithm during the design of a program.

(a) One part of the program contains an algorithm which is represented by a state-transition diagram.

The table shows the inputs, outputs and states for the algorithm:

Current state	Input	Output	Next state
S1	A1		S2
S2	A2	X4	S3
S3	A1	X1	S3
S3	A2	X1	S3
S3	A3	X3	S4
S3	A4		S2
S4	A3	X4	S5
S4	A4	X4	S5
S4	A1		S2

Complete the state-transition diagram to represent the information given in the table:



[5]

(b) A structure chart is used to document a different part of the program, made up of five modules.

Program notes:
- module Setup calls either module Restart, or module Confirm
- module Confirm takes a string as a parameter and returns an integer
- module Modify has no parameters and returns a Boolean
- module Update takes a string as a parameter
- module Restart repeatedly calls module Update followed by module Modify
- module Restart takes a string as a parameter that is passed by reference.

Draw a structure chart to represent the relationship between the **five** modules, including all parameters and return values.

[4]

5 A program is developed to satisfy a specific customer requirement.

The project follows a program development life cycle model. This model divides the development process into several different stages.

(a) The table lists some of the development activities.

Complete the table by writing the name of the life cycle stage for each activity:

Activity	Name of life cycle stage
a structure chart is produced	
a program is modified to allow it to run on new hardware	
the programmer identifies the customer's requirements	
an Integrated Development Environment (IDE) provides context-sensitive help such as 'auto-complete'	

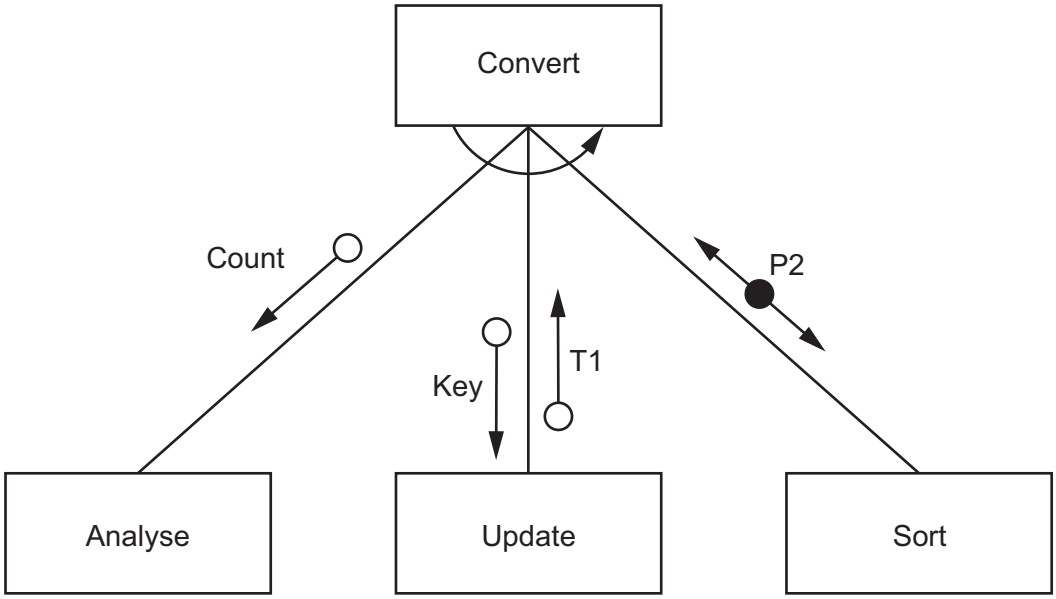
[4]

(b) Alpha and beta testing has been completed. The final testing stage is carried out by the customer.

Identify this final testing stage.

..... [1]

7 The structure chart shows part of a program design:



(a) Explain the meaning of the curved arrow symbol in this structure chart.

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..... [2]

(b) The program designer has noted that:

- Count is the number of elements in an array
- T1 is a value representing the number of elements in an array that have been modified
- Key is of type `STRING`

Write the pseudocode module headers for analyse, update and sort.

Analyse:

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Update:

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Sort:

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[4]

8 A program is being developed to manage student book loans from a college library.

The programmer has defined a record type to define each loan.

The data items are:

Data item	Data type	Comment
`StudentID`	`STRING`	the unique ID of the student who has borrowed the book
`BookID`	`STRING`	the unique ID of the book being borrowed
`OnLoan`	`BOOLEAN`	`TRUE` if the book has **not** been returned

The programmer has defined a global array `Loan` to store 7000 loan records.

There are more elements in the array than books in the library. Unused elements have the `StudentID` set to an empty string. These may occur anywhere in the array.

The programmer has defined a program module:

Module	Description
`CountLoans()`	• called with a parameter of type `STRING` representing a `StudentID` • counts the number of books currently on loan to the specified student • counts the number of books that the student has already returned • output both counts together with a suitable message

(a) Write pseudocode for module `CountLoans()`

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(d) It is decided to extend the program so that a new module `Reminder()` will send an email to the student three days before the book is due to be returned.

Outline the changes that will need to be made to the data stored and how this data will be used to generate the reminder email.

Data .....

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Use .....

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[2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program is being developed to meet a particular customer requirement.

(a) The program contains these variables:

Variable	Data type
MyChar	CHAR
MyString	STRING
MyInt	INTEGER
MyDOB	DATE

Complete the table by filling in the gaps using functions and/or operators from the **insert**.

Each expression must be valid.

Expression
MyString ← 'X' ..... MyString
MyChar ← ..... ("ABCD", ... ,.....)
MyString ← ..... (..... (MyDOB))
MyInt ← ..... (..... (MyString) / 2)

[4]

(b) Different test methods will be used at different stages of the program development.

Complete the table by identifying the test method that matches the test description.

The first row has been completed for you.

Test description	Test method
carried out as soon as a program module has been coded	alpha
carried out as program modules are being combined	
completed by the developers without referring to the code	
completed by the customer	

[3]

(c) During the alpha testing stage, an Integrated Development Environment (IDE) is used to help locate an error that has been identified. The IDE report window feature is used to examine the values assigned to variables.

Explain how **two** other IDE features are used together with the report window feature to help locate the error.

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..... [3]

6 Students are learning about a simple check digit method for data validation. In this method, a single check digit is appended to the end of an original number to give a new number.

The students are studying a method which:

- calculates the sum of all the digits in the original number
- uses integer division to calculate the remainder when the sum is divided by 10
- uses the remainder as the check digit
- appends the check digit to the original number, creating the new number.

For example:

original number	4162
sum of all digits	4 + 1 + 6 + 2 = 13
remainder when the sum is divided by 10 using integer division	3
new number	41623

The method described can be used to detect single-digit errors. For example, if the new number is incorrectly input as 41633, then the check digit does not match and the input will be rejected.

(a) An incorrect attempt was made to enter 41623. The **first two** digits were entered incorrectly; the **last three** digits were entered correctly.

When this number was tested, the check digit was found to be correct and the input was accepted.

Identify an example of the incorrect attempt to enter 41623 and explain why this number would **not** be rejected.

Number .....

Explanation .....

.....

[2]

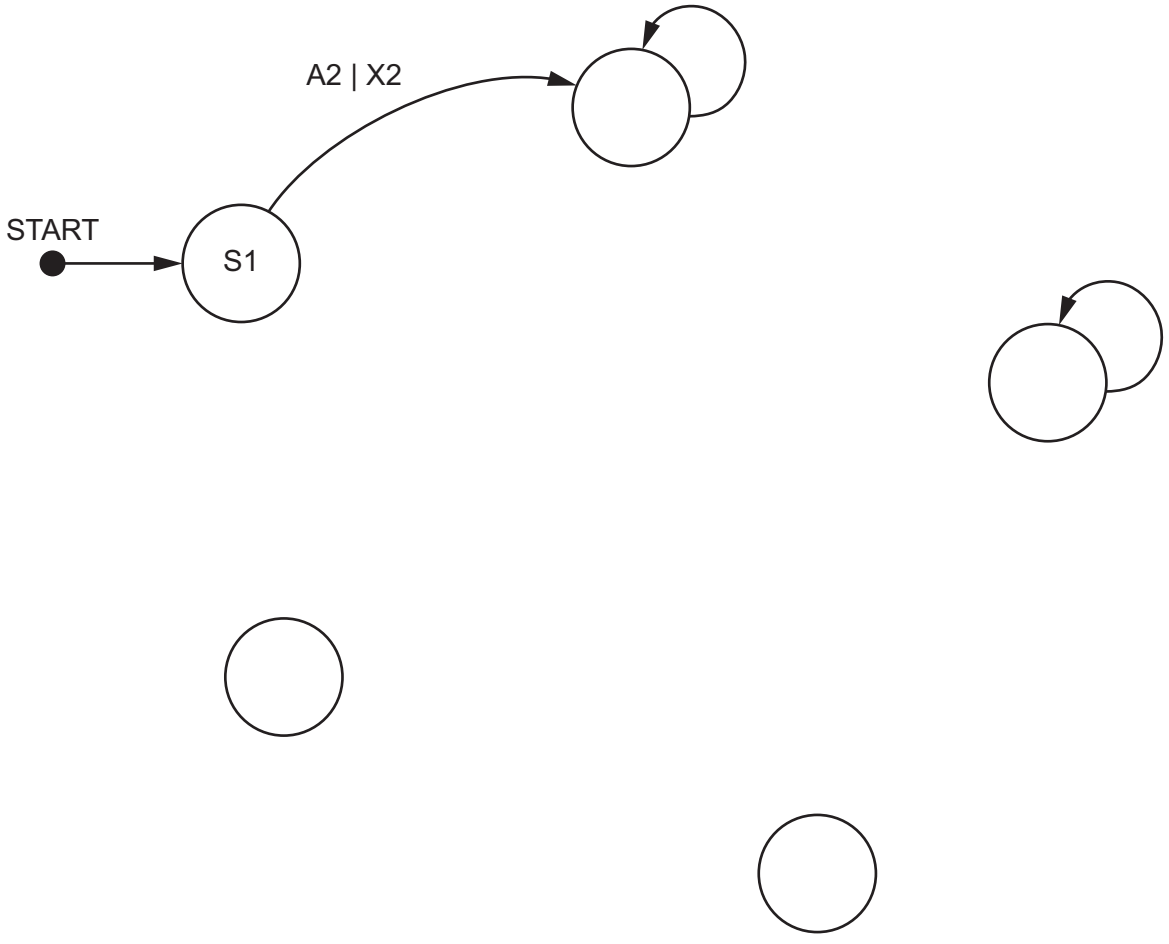
7 There are several different ways to express an algorithm during the design of a program.

(a) One part of the program contains an algorithm which is represented by a state-transition diagram.

The table shows the inputs, outputs and states for the algorithm:

Current state	Input	Output	Next state
S1	A2	X2	S3
S1	A1	X1	S2
S2	A4	X4	S5
S3	A1		S3
S3	A3	X3	S2
S3	A2	X4	S4
S4	A1	X1	S4
S4	A3		S2
S4	A4	X4	S5

Complete the state-transition diagram to represent the information given in the table.



[5]

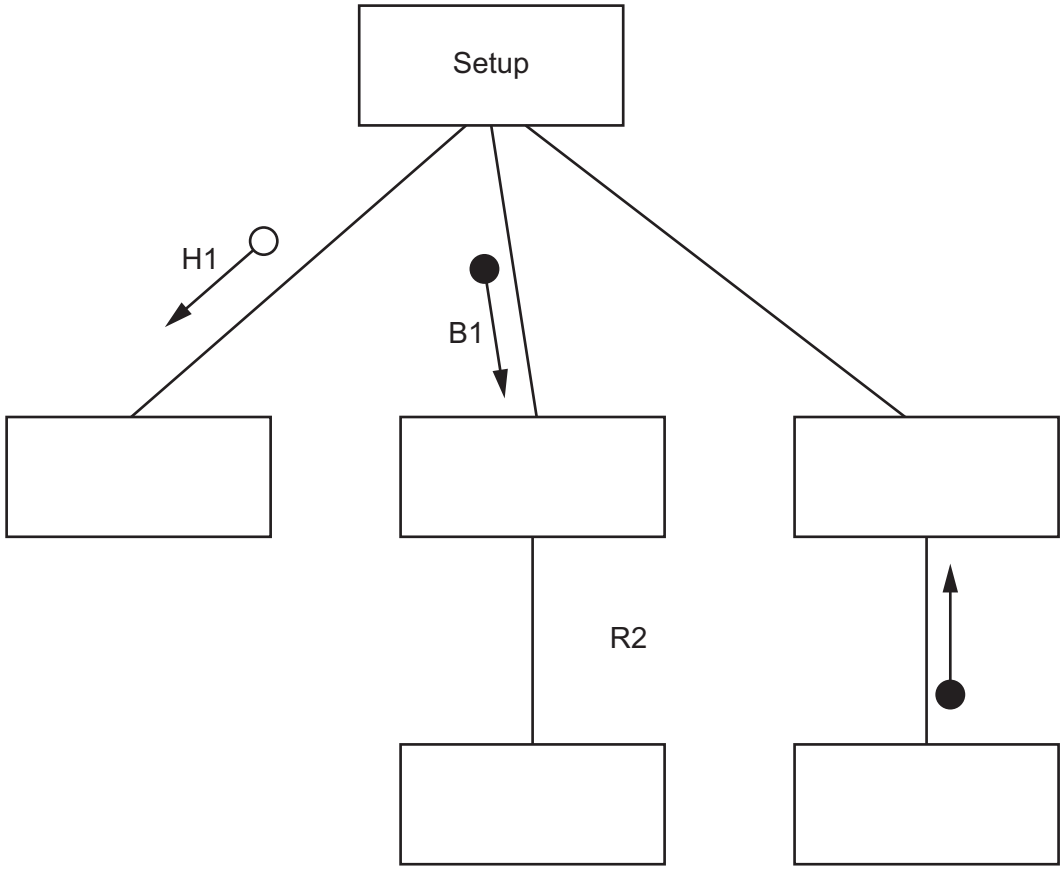
(b) A structure chart is used to document a different part of the program.

This part of the program contains six modules:

Pseudocode module header
`PROCEDURE Setup()`
`PROCEDURE Restart(H1 : STRING, C1 : INTEGER)`
`FUNCTION Modify(B1 : BOOLEAN) RETURNS INTEGER`
`PROCEDURE Final(T1 : INTEGER)`
`PROCEDURE Update(BYREF R2 : STRING)`
`FUNCTION Confirm() RETURNS BOOLEAN`

Module Setup will repeatedly call **three** of the modules.

Complete the structure chart to document the information given in the table.



[5]

8 A program is being developed to manage student book loans from a college library. Students may borrow up to five books at a time from the library.

The programmer has defined a record type to define each loan.

The record data items are:

Data item	Data type	Comment
`StudentID`	`STRING`	the unique ID of the student who has borrowed the book
`BookID`	`STRING`	the unique ID of the book being borrowed
`OnLoan`	`BOOLEAN`	`TRUE` if the book has **not** been returned

The programmer has defined a global array `Loan` to store 5000 loan records.

There are more elements in the array than books in the library. Unused elements have the `StudentID` set to an empty string. These may occur anywhere in the array.

The programmer has defined the first program module:

Module	Description
`OKToBorrow()`	• called with a parameter of type `STRING` representing a `StudentID` • search the array for loan records for the specified student • output a suitable message to say whether the student may, or may not, borrow another book

(a) Write efficient pseudocode for the module `OKToBorrow()`

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(c) It is decided to introduce a system of fines for books that have been borrowed for too long.

Two new requirements are defined:

1. Each loan has a maximum length, represented as a number of days.
2. Each book in the library will be assigned one of three categories. Each category has a different maximum loan length.

Record structure and program design changes are needed to meet these two requirements.

Outline the changes that are necessary to meet the **two** requirements.

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..... [2]

4 A programmer is developing a new stock control program for a shop owner to produce sales reports.

(a) A text file `Sales.txt` stores strings representing the number of items sold.

The file contains 52 lines, each representing the number of items sold in each week of the year.

For example:

File line number	File data
1	"350"
2	"502"
3	"434"
49	"492"
50	"872"
51	"772"
52	"201"

The example shows that 502 items were sold in week 2.

The owner requires a module to output all week numbers where the number of items sold is greater than 500.

Describe an algorithm that produces the required module.

Do **not** include pseudocode statements in your answer.

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..... [5]

(b) Once the program has been completed, it is tested using the walkthrough method.

Describe the walkthrough method and explain how it can be used to identify errors.

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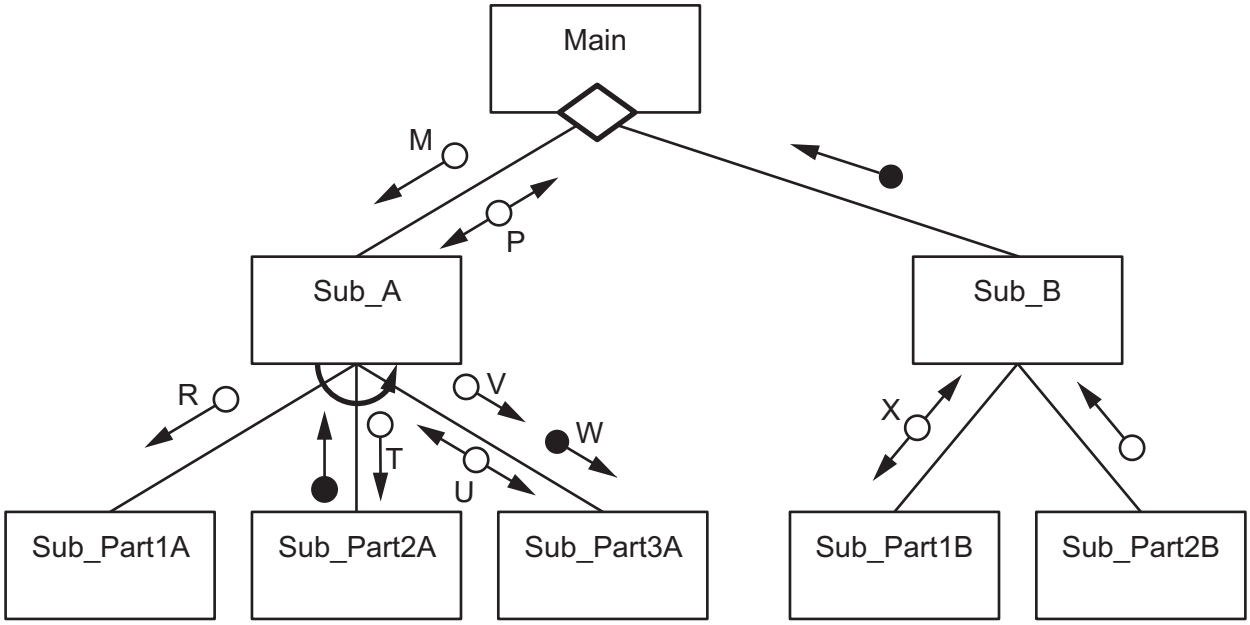
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..... [3]

6 Study the structure chart:



Some of the parameter data types are:

- R and V are of type `INTEGER`
- T is of type `REAL`
- U is of type `STRING`
- W is of type `BOOLEAN`.

Some of the modules in the structure chart are functions, others are procedures.

(a) Explain **one** difference between functions and procedures.

.....

..... [1]

(b) Write the pseudocode to define the module headers:

Sub_Part1A

.....

.....

Sub_Part2A

.....

.....

Sub_Part3A

.....

.....

[5]

(c) The structure chart uses the two symbols shown.

Complete the table to explain what each symbol means and how the relevant modules in the structure chart are affected.

Symbol	Explanation
	....
	....

[3]

7 A program is being developed to implement a customer loyalty scheme for a coffee shop.

The programmer has decided that the following data items need to be stored for each customer:

Data item	Description
customer ID	a unique six-digit string
loyalty points	an integer value that depends on how often the customer visits the coffee shop
last visit date	the date the customer last visited the coffee shop

The programmer has defined a program module:

Module	Description
`UpdateVisit()`	• called with two parameters: 1. loyalty points of type `INTEGER` 2. last visit date of type `DATE` • change the loyalty points: ○ increase by **four** if the last visit date and the current date are in the same month ○ otherwise increase by **one** • change the last visit date to the current date • the changed values must be available to the code that follows the call to `UpdateVisit()`

(a) (i) Write pseudocode for module `UpdateVisit()`

Assume the customer has made at least **one** previous visit.

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.... [5]

(ii) The programmer decides to amend the `UpdateVisit()` module so that loyalty points are further increased if the customer visits the coffee shop the same day of the week as their last visit.

Write the pseudocode for this condition.

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..... [1]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program is being developed to control the production line in a factory.

(a) A number of different program life cycles are available for the development of a program.

(i) Explain the need for different program development life cycles.

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..... [1]

(ii) Coding is a stage in a program development life cycle.

State **one** consideration that would influence the choice of programming language.

.....

..... [1]

(b) The program has been in use for a number of months and adaptive maintenance is required.

(i) Give **three** reasons why adaptive maintenance may be required.

1 .....

2 .....

3 .....
[3]

(ii) As well as adaptive maintenance, other types of program maintenance may be needed.

Identify **one** other type of program maintenance.

..... [1]

2 A program to calculate the pay of employees working for a company is being designed.

(a) Stepwise refinement has been used in the design of the program.

Describe stepwise refinement.

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..... [2]

(b) One of the program modules used to calculate employees' weekly bonus pay has been completed. The amount of bonus pay they receive is based on the number of hours worked and the value of sales made as shown in the table.

Bonus pay and value of sales are both in dollars.

Hours worked	Value of sales	Bonus pay
between 1 and 40 inclusive	2000 or less	0
between 1 and 40 inclusive	above 2000	50
above 40	2000 or less	10
above 40	above 2000	100

(i) The module is tested using white-box testing.

State **two** tests, using valid data, that can be used to test different paths through the program.

Test one:

Hours worked .....

Value of sales .....

Bonus pay .....

Test two:

Hours worked .....

Value of sales .....

Bonus pay .....

[2]

(ii) The program to calculate pay uses a number of modules which are each called from different places in the program.

The program is to be tested using stub testing before all the program modules have been completed.

Describe stub testing.

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..... [2]

(iii) The program has been completed and compiles successfully.

The program is tested using black-box testing.

Identify and describe **one** type of error that black-box testing could detect.

Type of error .....

Description .....

.....

.....

[2]

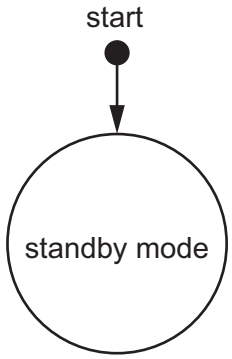
5 An automated digital camera system is used to take a sequence of pictures of animals in a garden. During the design of the system, a state-transition diagram is produced.

The table details the states in the automated digital camera system along with the events which cause the states to change.

The system starts in standby mode. The sequence of pictures is taken when in active mode.

Current state	Event	Next state
standby mode	turn on	detect mode
detect mode	turn off	standby mode
detect mode	movement detected	active mode
active mode	20 seconds elapsed	sequence complete
active mode	turn off	standby mode
sequence complete	time saved	detect mode

(a) Complete the state-transition diagram for the automated digital camera system.



[4]

6 Two arrays `Data` and `Pointer` are accessed by the procedure `Place()`

`Data` and `Pointer` are both global arrays of type `INTEGER`

The contents of these two arrays are shown:

	Data
1	1018
2	1007
3	1010
4	1056
5	1092
6	1062
7	1034
8	0
9	0
10	0

	Pointer
1	7
2	3
3	1
4	6
5	-1
6	5
7	4
8	9
9	10
10	-1

Study the pseudocode:

```
PROCEDURE Place(Value : INTEGER, Start : INTEGER, Unused : INTEGER)
   DECLARE New, Current, Last : INTEGER
   CONSTANT NullPointer = -1
   New ← Unused
   Last ← NullPointer
   Current ← Start
   WHILE Current <> NullPointer AND Data[Current] < Value
      Last ← Current
      Current ← Pointer[Current]
   ENDWHILE
   Pointer[New] ← Pointer[Last]
   Pointer[Last] ← New
   Data[New] ← Value
ENDPROCEDURE
```

(a) (i) Complete the trace table below by dry running the procedure `Place()` when it is called by the statement:

```
CALL Place(1043, 2, 8)
```

Value	Start	Unused	New	Last	Current

[4]

(ii) Complete the diagram showing the contents of the global arrays `Data` and `Pointer` after the procedure `Place()` has run to completion when called as shown in part **(a)(i)**.

	Data
1	1018
2	1007
3	1010
4	1056
5	1092
6	1062
7	1034
8	
9	
10	

	Pointer
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	

[3]

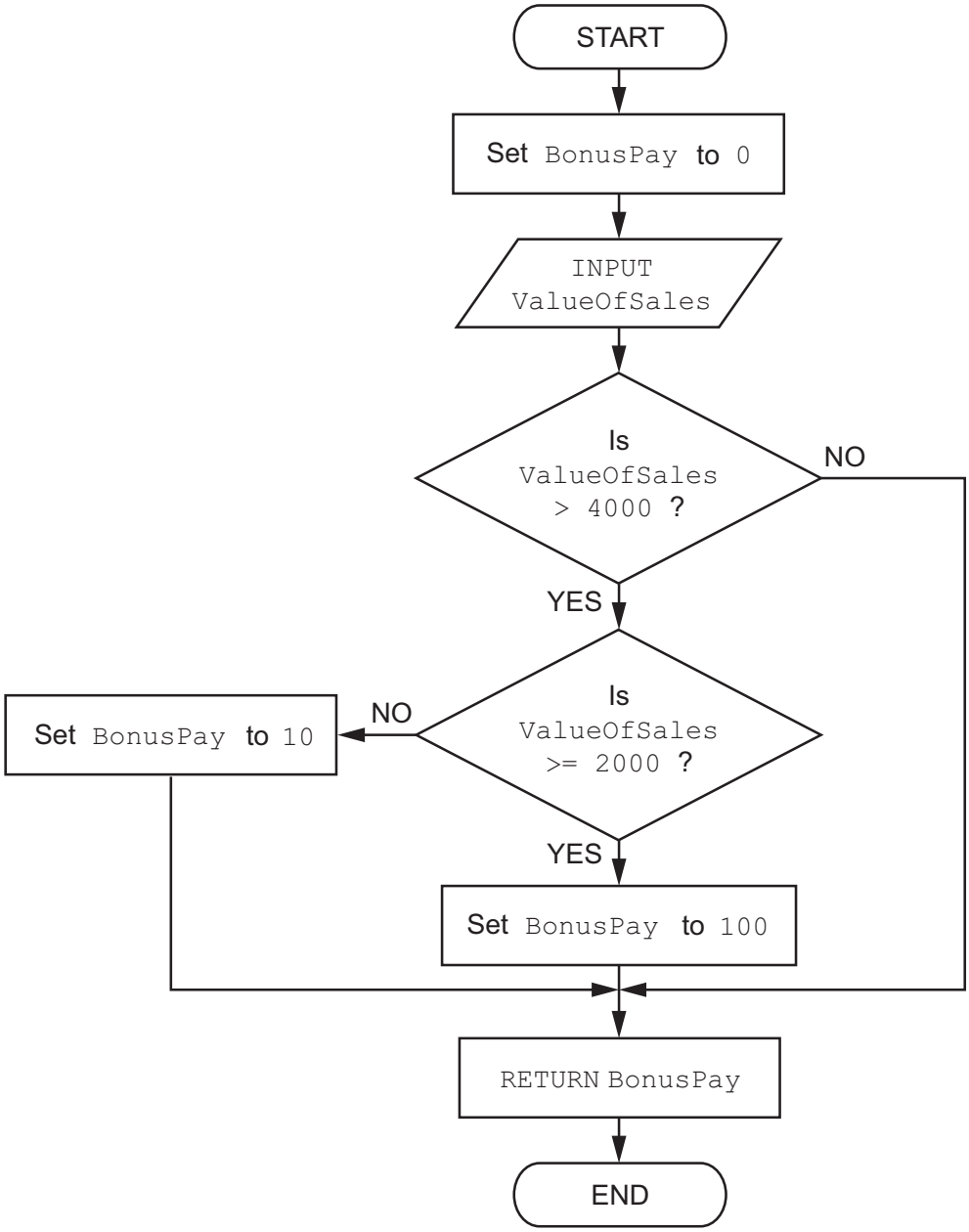
2 A program is being developed to calculate the pay of employees working for a company.

A function `CalculateBonus()` calculates bonus pay based on the value of sales.

Bonus pay is calculated as shown in the table.

Value of sales (in dollars)	Bonus pay (in dollars)
below 2000	0
between 2000 and 4000 inclusive	10
above 4000	100

A flowchart for the function `CalculateBonus()` has been designed.



(a) The flowchart contains logic errors.

One logic error is that `BonusPay` will **not** be set to `10` although the `ValueOfSales` input is between `2000` and `4000` inclusive.

(i) Explain why this error occurs.

.....

..... [1]

(ii) Explain how this error could be corrected.

.....

..... [1]

(b) There are different ways to reduce the risk of errors when developing the new program, such as the use of constants.

(i) State a value that could be replaced by a constant in the function `CalculateBonus()`

..... [1]

(ii) Explain how the use of constants helps to reduce the risk of programming errors.

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..... [2]

(iii) One other way that can reduce the risk of errors when writing the program is the use of library routines.

Explain how library routines can reduce the risk of programming errors.

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..... [1]

(c) All the logic errors have been corrected, and the program has been coded.

Identify and describe **two** other types of error that the program could contain.

Type of error .....

Description .....

.....

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Type of error .....

Description .....

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.....

[4]

4 Study the algorithm:

```
DECLARE Chars : ARRAY[1:4] OF CHAR
DECLARE I, J : INTEGER
DECLARE Key : CHAR
I ← 2
Chars[1] ← 'D'
Chars[2] ← 'T'
Chars[3] ← 'H'
Chars[4] ← 'R'
WHILE I <= 4                          //Outer loop
   Key ← Chars[I]
   J ← I - 1
   WHILE J >= 0 AND Chars[J] > Key     //Inner loop
      Chars[J + 1] ← Chars[J]
      J ← J - 1
   ENDWHILE
   Chars[J + 1] ← Key
   I ← I + 1
ENDWHILE
```

(a) The outer loop structure used in the algorithm is **not** the most appropriate one to use.

State the type of loop structure that would be the most appropriate to use and justify why it is the most appropriate.

Loop structure .....

Justification .....

.....

[2]

(b) Complete the trace table by dry running the algorithm.

The first row has been completed.

I	Key	J	Chars[J]	Chars			
				[1]	[2]	[3]	[4]
2				'D'	'T'	'H'	'R'

[6]

6 A program monitors the speed of vehicles as they move around a large building site.

Each vehicle contains a sensor which reads an integer value that represents the speed of the vehicle. The value is expected to be in the range 0 to 60 inclusive.

The sensors cannot read values less than 0.

A program module has been written to validate the values read by the sensors.

(a) A test plan is needed to fully test the module.

Complete the table. The first line has been completed for you.

Assume the sensors generate only integer values.

Type of test data	Test data value	Expected outcome
normal	36	data item is accepted

[4]

7 A program is being developed to implement a customer loyalty scheme for a coffee shop.

The programmer has decided that the following data items need to be stored for each customer:

Data item	Description
customer ID	a unique six-digit string
points	an integer value that is increased by one for every cup of coffee the customer orders

When a customer visits the shop and orders coffee, the scheme operates as follows:

- The total number of points is increased by the number of coffees ordered.
- If just one cup of coffee is ordered and the number of points goes above 10, then:
  - the cup of coffee they have just ordered is given to them free of charge
  - the number of points is reduced by 11.
- If the order is for multiple coffees and the number of points goes above 10, then:
  - they get one coffee free of charge for every 11 points
  - the number of points is reduced by 11 for each free coffee.

For example, the:

- customer currently has 9 points
- customer orders 16 cups of coffee
- total number of points now becomes 25
- customer gets 2 free coffees and now has 3 points left.

The programmer has defined a program module that is called every time a customer places an order:

Module	Description
`CustomerOrder()`	• called with two integer parameters: ◦ the number of coffees ordered ◦ the current number of points • output a suitable message giving the number of free coffees • return a value for the new points total.

(a) Write pseudocode for module `CustomerOrder()`

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..... [6]

(b) A text file `Loyalty.txt` will be used to store the data items for the loyalty scheme. The data items for each customer will be stored on a separate line of the text file where each data item is separated by a comma:

```
<CustomerID>,<Points>
```

The contents of the text file `Loyalty.txt` will always be stored in ascending order by customer ID.

When the data items are read from or written to the text file `Loyalty.txt`, they may need to be converted to the appropriate data type.

Each customer has a unique customer ID starting at `"100001"` with this value increasing by one each time a new customer joins the loyalty scheme.

When a customer joins the loyalty scheme, they are assigned the next customer ID and value of points is set to 0.

For example, if the loyalty scheme has 204 customers and a new customer joins the loyalty scheme, the following line is added to the text file `Loyalty.txt`:

```
"100205,0"
```

You can assume that the number of customers in the loyalty scheme will never be more than 9000.

The programmer has defined a program module as follows:

Module	Description
`AddNewCustomers()`	• called with an integer parameter representing the number of new customers to be added to the loyalty scheme • adds a new line, containing the required information, to the text file `Loyalty.txt` for each customer added to the loyalty scheme • outputs each new customer ID added to the loyalty scheme

(i) Write pseudocode for module `AddNewCustomers()`

Assume that there is at least **one** customer already in the loyalty scheme.

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(ii) The requirements for `AddNewCustomers()` are changed. There may be no existing customers in the loyalty scheme.

Explain the changes that will need to be made to the module `AddNewCustomers()`

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..... [4]

5 A software developer follows a program development life cycle. The life cycle divides the development process into various stages.

(a) The following table lists some development activities.

Complete the table by writing the name of the life cycle stage for each activity.

Activity	Name of life cycle stage
A compiler is used.	
A program that has been released for general use is modified.	
The dry run method is used.	
The program structure is specified.	

[4]

(b) A software developer has written modules `Test_A()` and `Test_B()`. These have been written but contain errors. These modules are called from several places in the main program and testing of the main program (integration testing) has to stop.

Identify a method that can be used to continue testing the main program **before** the errors in these modules have been corrected **and** describe how this would work.

Method .....

Description .....

.....

.....

.....

[3]

7 A coffee shop owner wants to introduce a computerised loyalty card system.

A programmer discusses the details of the system with the shop owner.

(a) Identify the stage of the program development life cycle that this discussion is part of **and** give a document that will be produced during this stage.

Stage .....

Document .....

.....

[2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program has been developed and released for general use. After a few months of use an error is detected where under certain circumstances it outputs an unexpected value.

(a) The error in the program needs to be corrected.

Identify the stage of the program development life cycle that this correction is made in.

..... [1]

(b) The program contains a function `Lookup()`. After investigation, it is found that this is the function that sometimes returns an incorrect value.

An Integrated Development Environment (IDE) is used to help locate the error.

The IDE features of watch window, single stepping and breakpoint will be used.

Explain these features including the order that they will be used in to locate the error in `Lookup()`.

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..... [3]

(c) To solve the error a programmer decides to create a new module.

The design of the new module has been completed and the module is being coded.

Identify **two** features of an IDE that will help during the coding of this new module.

1 .....

2 .....

[2]

5 A program contains a global 1D array `Data` with 100 elements of type `INTEGER`.

The program contains a function `Process()` expressed in pseudocode as follows:

```
FUNCTION Process(Number : INTEGER, Label : STRING) RETURNS STRING
   DECLARE Index, Count : INTEGER
   DECLARE ReturnValue : STRING

   Count ← INT(100 / Number)
   Index ← Data[Number]

   CASE OF (Index MOD 2)
      0 : ReturnValue ← TO_UPPER(RIGHT(Label, Count))
      1 : ReturnValue ← "****"
   ENDCASE

   RETURN RetVal
ENDFUNCTION
```

(a) Run-time errors can be generated in different ways. For example, a run-time error will be generated if a function is called with invalid parameters.

The pseudocode contains three statements that could generate a run-time error.

Write the **three** statements **and** explain how each could generate a run-time error.

Statement 1 .....

Explanation .....

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Statement 2 .....

Explanation .....

.....

Statement 3 .....

Explanation .....

.....

[3]

(b) One type of run-time error can cause a program to stop responding ('freezing').

Identify a particular type of programming construct that can generate this type of error **and** explain why it occurs.

Construct .....

Explanation .....

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.....

[2]

6 A shop sells sandwiches and snacks. The owner chooses a 'daily special' sandwich which is displayed on a board outside the shop. Each 'daily special' has two different fillings and is made with one type of bread.

The owner wants a program to randomly choose the 'daily special' sandwich.

The program designer decides to store the possible sandwich fillings in a 1D array of type string.

The array is declared in pseudocode as follows:

```
DECLARE Filling : ARRAY [1:35] OF STRING
```

Each element contains the name of one filling.

An example of the first five elements is as follows:

Index	Element value
1	`"Cheese"`
2	`"Onion"`
3	`"Salmon"`
4	`"Anchovies"`
5	`"Peanut Butter"`

A second 1D array stores the possible bread used:

```
DECLARE Bread : ARRAY [1:10] OF STRING
```

Each element contains the name of one type of bread.

An example of the first three elements is as follows:

Index	Element value
1	`"White"`
2	`"Brown"`
3	`"Pitta"`

Both arrays may contain unused elements. The value of these will be an empty string and they may occur anywhere in each array.

A procedure `Special()` will output a message giving the 'daily special' sandwich made from **two** randomly selected different fillings and **one** randomly selected bread.

Unused array elements must **not** be used when creating the 'daily special' sandwich.

Using the above examples, the output could be:

```
"The daily special is Cheese and Onion on Brown bread."
```

(a) Complete the pseudocode for the procedure `Special()`.

Assume that both arrays are global.

```
PROCEDURE Special()
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ENDPROCEDURE
```

[7]

(b) The owner decides that some combinations of fillings do not go well together. For example, anchovies and peanut butter.

Describe how the design could be changed to prevent certain combinations being selected.

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..... [2]

7 A coffee shop runs a computerised loyalty card system.

Customers are issued with a loyalty card with their name together with a unique customer ID.

Loyalty points are added to their card each time they spend money at the shop.

The following information is stored for each customer: ID, name, home address, email address, mobile phone number, date of birth, number of points, date of last visit and amount of money spent at last visit.

A new module will generate a personalised email message to each loyalty card customer who has not visited the coffee shop in the last three months. The message will include a unique voucher code which can be used to authorise a discount if the customer goes to the shop within the next two weeks.

(a) Identify **three** items of customer information that will be used by the new module **and** justify your choices.

Item 1 .....

Justification .....

.....

Item 2 .....

Justification .....

.....

Item 3 .....

Justification .....

.....

[3]

(c) It is decided to adopt a formal program development life cycle model for the development of the new module.

The analysis of the new module is complete and the project moves on to the design stage. During this stage all the necessary algorithms and module designs will be defined.

State **three other** items that will be defined for the new module during the design stage.

1 .....

2 .....

3 .....

[3]

(d) Part of the coffee shop program contains three program modules as follows:

- Module `Init()` has no parameters and returns a Boolean.
- Module `Reset()` takes a string as a parameter and returns an Integer.
- Module `Check()` repeatedly calls `Init()` followed by `Reset()`.

Draw a structure chart to represent the relationship between the **three** modules, including all parameters and return values.

[3]

BLANK PAGE

8 A program is being developed to implement a game for up to six players.

During the game, each player assembles a team of characters. At the start of the game there are 45 characters available.

Each character has four attributes, as follows:

Attribute	Examples	Comment
Player	0, 1, 3	The player the character is assigned to.
Role	Builder, Teacher, Doctor	The job that the character will perform in the game.
Name	Bill, Lee, Farah, Mo	The name of the character. Several characters may perform the same role, but they will each have a unique name.
Level	14, 23, 76	The skill level of the character. An integer in the range 0 to 100 inclusive.

The programmer has defined a record type to define each character.

The record type definition is shown in pseudocode as follows:

```
TYPE CharacterType
   DECLARE Player : INTEGER
   DECLARE Role : STRING
   DECLARE Name : STRING
   DECLARE Level : INTEGER
ENDTYPE
```

The `Player` field indicates the player to which the character is assigned (1 to 6). The field value is 0 if the character is **not** assigned to any player.

The programmer has defined a global array to store the character data as follows:

```
DECLARE Character : ARRAY[1:45] OF CharacterType
```

At the start of the game all record fields are initialised, and all `Player` field values are set to 0

The programmer has defined a program module as follows:

Module	Description
`Assign()`	• called with two parameters: ○ an integer representing a player ○ a string representing a character role • search the `Character` array for an unassigned character with the required role • If found, assign the character to the given player and output a confirmation message, for example: `"Bill the Builder has been assigned to player 3"` • If no unassigned character with the required role is found, output a suitable message.

(a) Write pseudocode for module `Assign()`.

[7]

(b) A new module will store the contents of the `Character` array in a text file.

The module is defined as follows:

Module	Description
`Save()`	• form a string from each record with fields separated by the character `'^'` • write each string to a separate line of the new file named `SaveFile.txt`

Complete the pseudocode for module `Save()`.

```
PROCEDURE Save()
```

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```
ENDPROCEDURE
```

[7]

(d) The save operation is to be extended so that multiple files may be saved as the game progresses. This will allow the user to restore the game from any saved position. The filename must reflect the sequence in which the files are saved.

Describe a method that would allow multiple files to be saved **and** give an example of **two** consecutive filenames.

Method .....

.....

Example .....

.....

[2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program will calculate the tax payable based on the cost of an item.

Calculations will occur at many places in the program and these involve the use of one of three tax rates.

Tax rate values represent a percentage. For example, a tax rate value of 5.23 represents 5.23%. In this case, the tax payable on an item costing $100 would be $5.23.

Tax rate values are used at several places within the program. One example is given in pseudocode as follows:

```
HighRate ← FALSE
CASE OF ItemCost
   <= 50  : TaxRate ← 3.75      // tax rate of 3.75%
   <= 200 : TaxRate ← 5.23      // tax rate of 5.23%
   > 200  : TaxRate ← 6.25      // tax rate of 6.25%
            HighRate ← TRUE
ENDCASE
TaxPayable ← ItemCost * TaxRate // tax payable
```

(a) The pseudocode contains a logical error.

Identify the error **and** suggest a correction.

Error .....

.....

Correction .....

.....

[2]

BLANK PAGE

3 An algorithm will output the **last** three lines from a text file `Result.txt`

The lines need to be output in the same order as they appear in the file.

Assume:

- Three variables `LineX`, `LineY` and `LineZ` will store the three lines. These are of type string and all three variables have been initialised to an empty string.
- The file exists and contains **at least** three lines.

(a) The algorithm to output the lines is expressed in eight steps.

Complete the steps.

1. Open the file .....
2. Loop until .....
3. ..... and store in `ThisLine`
4. Assign `LineY` to `LineX`
5. Assign `LineZ` to `LineY`
6. Assign `ThisLine` to `LineZ`
7. After the loop, ...
8. Output `LineX`, `LineY`, `LineZ`

[4]

(c) The requirement changes, and the algorithm will now output three lines from the file, starting from a **given** line number.

The modified algorithm will be implemented as a function which will:

- be called with an integer parameter representing the given line number
- output three lines, starting at the given line
- return `TRUE` if the 3 lines are output, or `FALSE` if it was not possible to output the 3 lines.

Describe the changes that need to be made to **steps 2 to 8** of the algorithm given in part **(a)**.

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..... [4]

4 A program includes the following assignment statement:

```
Result ← STR_TO_NUM(x) / STR_TO_NUM(y)
```

When the program evaluates the expression in the statement, it performs a calculation.

Variable `Result` is of type real and variables `x` and `y` are of type string.

Two checks are required before the calculation is performed:

1. The two strings represent valid numeric values.
2. The numeric value of string `y` is not zero.

(a) Identify the type of error that could occur if these checks are **not** carried out **and** state a cause of this error.

Type .....

.....

Cause .....

.....

[2]

5 A software developer follows a program development life cycle. The life cycle divides the development process into various stages.

(a) The following table lists some development activities.

Complete the table by writing the name of the life cycle stage for each activity.

Activity	Name of life cycle stage
The walkthrough method is used.	
An algorithm is implemented in a programming language.	
The client is interviewed about problems with the current system.	
The program is modified to run on new hardware.	
Records and file structures are defined.	

[5]

(b) The program contains a validation function.

(i) The function will:

- take an integer value as a parameter
- return `TRUE` if the value is within the range 24 to 37, inclusive
- otherwise return `FALSE`.

Complete the table to define a test plan to thoroughly test the operation of the function.

Type of test data	Test data value	Expected result
Normal	30	TRUE

[4]

(ii) The function is to be tested on its own. When it is shown to work correctly the function will be combined with other modules and testing will continue.

Identify the type of testing that this represents.

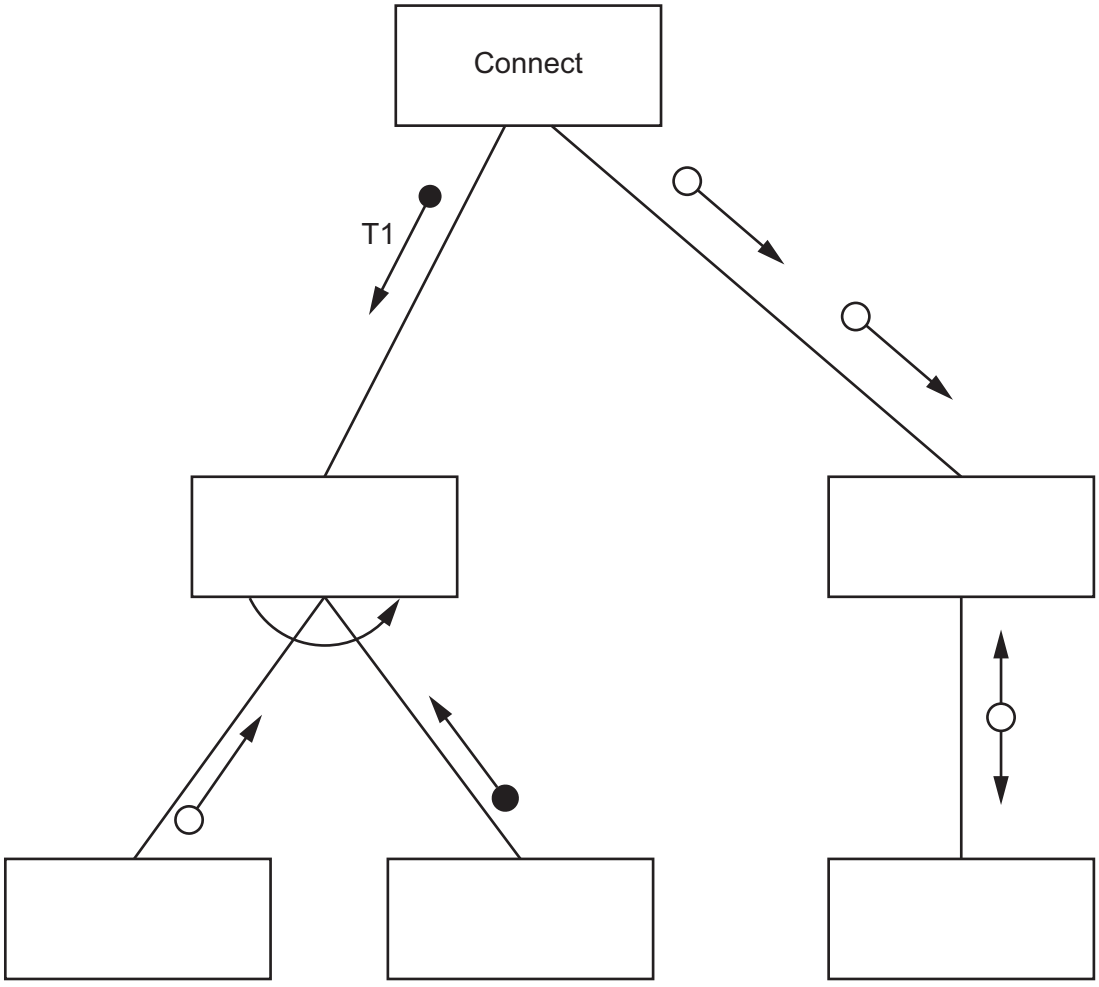
..... [1]

7 A program contains six modules with headers as follows:

Pseudocode module header
`PROCEDURE Connect()`
`FUNCTION Activate(H1 : STRING, Code : INTEGER) RETURNS BOOLEAN`
`FUNCTION Sync(T1 : BOOLEAN, S2 : REAL) RETURNS INTEGER`
`PROCEDURE Initialise(BYREF ID : INTEGER, BYVAL CC : INTEGER)`
`FUNCTION Reset(RA : STRING) RETURNS INTEGER`
`FUNCTION Enable(SA : INTEGER) RETURNS BOOLEAN`

Module `Connect()` will call either `Activate()` or `Sync()`. This is decided at run-time.

(a) Complete the structure chart for these modules.



[5]

(b) Explain the meaning of the curved arrow symbol used in the diagram in part **(a)**.

.....

..... [2]

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8 An exam paper has a maximum of 75 marks. One of five pass grades (A to E) is assigned, depending on the mark obtained. The lowest mark for a given grade is known as the grade boundary. For example, if the grade boundary for an A grade is 65 marks, then any candidate who achieves a mark of 65 or above will be awarded an A. A grade of U is awarded for marks below the E grade boundary.

The five grade boundaries are stored in a global 1D array `GradeBoundary` of type integer.

For example:

Element	Value	Comment
`GradeBoundary[1]`	65	The minimum mark for an A grade.

(b) A second module is defined:

Module	Description
`CheckAll()`	• called with a parameter of type integer representing the number of candidate marks in the `Result` array • uses `CheckMark()` to check each candidate mark • for each paper that needs to be checked, write the corresponding candidate ID on a separate line in a new file named `GRList.txt` • outputs a message with a count of how many papers need to be checked

Write pseudocode for module `CheckAll()`.

`CheckMark()` must be used to check each individual mark.

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..... [8]

(c) The requirement changes. Instead of a new file, the module described in part **(b)** needs to add the corresponding candidate ID for each paper that needs to be checked to an **existing** file.

Explain the change that will need to be made to `CheckAll()`.

.....

..... [1]

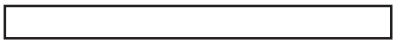
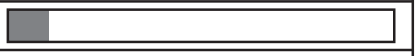
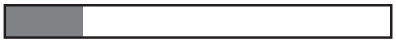
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6 A program displays a progress bar to inform the user of the progress of tasks that take a significant time to complete, such as those involving file transfer operations.

Task progress is divided into 11 steps. Each step represents the amount of progress as a percentage. An image is associated with each step and each image is stored in a different file.

Different progress bar images may be selected. For a given image, files all have the same filename root, with a different suffix.

The table illustrates the process for using the image with filename root `BargraphA`

Step	Percentage progress	Image filename	Image
1	< 10	`BargraphA-1.bmp`	
2	>= 10 and < 20	`BargraphA-2.bmp`	
3	>= 20 and < 30	`BargraphA-3.bmp`	
9	>= 80 and < 90	`BargraphA-9.bmp`	
10	>= 90 and < 100	`BargraphA-10.bmp`	
11	100	`BargraphA-11.bmp`	

A procedure `Progress()` will:

- be called with two parameters:
  - an integer representing the percentage progress (0 to 100 inclusive)
  - a string representing the image filename root
- generate the full image filename
- call a procedure `Display()` using the full image filename as the parameter.

(a) Write pseudocode for procedure `Progress()`.

[6]

(b) The definition of procedure `Progress()` is provided here for reference:

> A procedure `Progress()` will:
> - be called with two parameters:
> - an integer representing the percentage progress (0 to 100 inclusive)
> - a string representing the image filename root
> - generate the full image filename
> - call a procedure `Display()` using the full image filename as the parameter.

`Progress()` will be rewritten and a new module `Progress2()` produced with these requirements:

- an additional parameter of type integer will specify the total number of steps
- the image filename will be returned (procedure `Display()` will **not** be called from within `Progress2()`).

(i) Write pseudocode for the new module header.

.....

.....

..... [2]

(ii) State **one** benefit of increasing the number of steps.

.....

..... [1]

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7 Seven program modules form part of a program. A description of the relationship between the modules is summarised below. Any return values are stated in the description.

Module name	Description
Mod-A	calls Mod-B followed by Mod-C
Mod-B	• called with parameters Par1 and Par2 • calls either Mod-D or Mod-E, determined when the program runs • returns a Boolean value
Mod-C	• called with parameters Par1 and Par3 • Par3 is passed by reference • repeatedly calls Mod-F followed by Mod-G
Mod-D	called with parameter Par2
Mod-E	• called with parameter Par3 • returns an integer value
Mod-F	called with parameter Par3
Mod-G	• called with parameter Par3 • Par3 is passed by reference

Parameters in the table are as follows:
- Par1 and Par3 are of type string.
- Par2 is of type integer.

(a) (i) Identify the modules that would be implemented as functions.

..... [1]

(b) Draw a structure chart to show the relationship between the seven modules and the parameters passed between them.

[6]

Refer to the **insert** for the list of pseudocode functions and operators.

1 (a) The following table contains pseudocode examples.

Each example may contain statements that relate to one or more of the following:
- selection
- iteration (repetition)
- input/output.

Complete the table by placing **one or more** ticks (✓) in each row.

Pseudocode example	Selection	Iteration	Input/Output

`FOR Index ← 1 TO 10`
`  Data[Index] ← 0`
`NEXT Index`			

`WRITEFILE ThisFile, "****"`			
`UNTIL Level > 25`			

`IF Mark > 74 THEN`
`  READFILE OldFile, Data`
`ENDIF`			

[4]

(b) Program variables have data types as follows:

Variable	Data type
`MyChar`	`CHAR`
`MyString`	`STRING`
`MyInt`	`INTEGER`

Complete the table by filling in each gap with a function (from the **insert**) so that each expression is valid.

Expression
`MyInt ← ..... (3.1415926)`
`MyChar ← ..... ("Elwood", 3, 1)`
`MyString ← ..... ( ..... (27.509))`
`MyInt ← ..... ( ..... ("ABC123", 3))`

[4]

(c) The variables given in part **(b)** are chosen during the design stage of the program development life cycle.

The choices are to be documented to simplify program maintenance.

State a suitable way of documenting the variables **and** give **one** piece of information that should be recorded, in addition to the data type.

.....

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..... [2]

5 A program is being designed in pseudocode.

The program contains a global 1D array `Data` of type string containing 200 elements.

The first element has the index value 1.

A procedure `Process()` is written to initialise the values in the array:

```
PROCEDURE Process(Label : STRING)
    DECLARE Index : INTEGER
    Index ← 0
    INPUT Data[Index]
    WHILE Index < 200
        Index ← Index + 1
        CASE OF (Index MOD 2)
            0 : Data[Index] ← TO_UPPER(Label)
            1 : Data[Index] ← TO_LOWER(Label)
            OTHERWISE : OUTPUT "Alarm 1201"
        ENDCASE
    NEXT Index
    OUTPUT "Completed " & Index & " times"
ENDPROCEDURE
```

(a) (i) The pseudocode contains **two** syntax errors and **one** other error.

Identify the errors.

Syntax error 1 .....

.....

Syntax error 2 .....

.....

Other error .....

.....

[3]

(b) After correcting all syntax errors, the pseudocode is translated into program code which compiles without generating any errors.

When the program is executed it unexpectedly stops responding.

Identify the type of error that has occurred.

..... [1]

7 A fitness club has a computerised membership system.

The system stores information for each club member: name, home address, email address, mobile phone number, date of birth and exercise preferences.

Many classes are full, and the club creates a waiting list for each class. The club adds details of members who want to join a class that is full to the waiting list for that class.

When the system identifies that a space is available in one of the classes, a new module will send a text message to each member who is on the waiting list.

(a) Decomposition will be used to break the new module into sub-modules (sub-problems).

Identify **three** sub-modules that could be used in the design **and** describe their use.

Sub-module 1 .....

Use .....

.....

.....

.....

Sub-module 2 .....

Use .....

.....

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.....

Sub-module 3 .....

Use .....

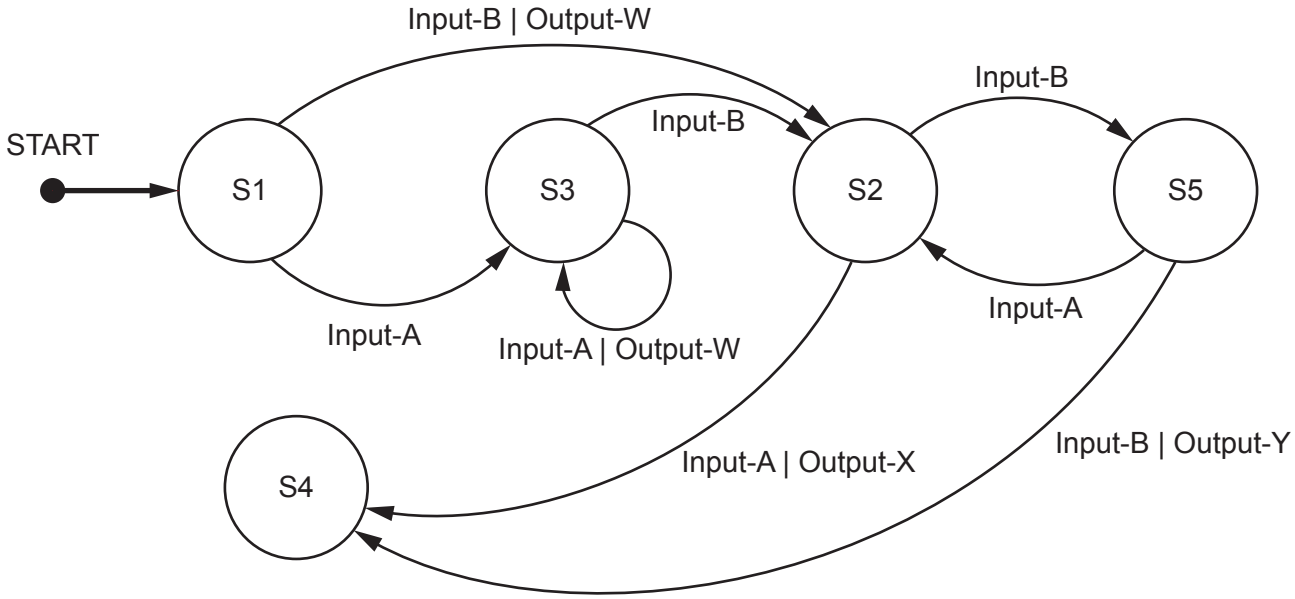
.....

.....

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[3]

(b) A different part of the program is represented by the following state-transition diagram.



(i) Complete the table to show the inputs, outputs and next states.

Assume that the current state for each row is given by the 'Next state' on the previous row. For example, the first Input-A is made when in state S1.

If there is no output for a given transition, then the output cell should contain 'none'.

The first two rows have been completed.

Input	Output	Next state
		S1
Input-A	none	S3
	Output-W	
	none	
Input-B		
Input-A		
		S4

[5]

(ii) Identify the input sequence that will cause the minimum number of state changes in the transition from S1 to S4.

..... [1]

Refer to the **insert** for the list of pseudocode functions and operators.

1 An algorithm is developed in pseudocode before being coded in a programming language.

(a) The following table shows four valid pseudocode assignment statements.

Complete the table by giving an appropriate data type to declare each of the variables `A`, `B`, `C` and `D`.

Assignment statement	Data type
`A ← LEFT(MyName, 1)`	
`B ← Total * 2`	
`C ← INT(ItemCost) / 3`	
`D ← "Odd OR Even"`	

[4]

(c) The variable names `A`, `B`, `C` and `D` in part **(a)** are **not** good programming practice.

(i) State why these variable names are **not** suitable.

.....

..... [1]

(iii) The choice of suitable variable names is one example of good programming practice.

Give **one other** example.

.....

..... [1]

5 A global 1D array of strings contains three elements which are assigned values as shown:

```
Data[1] ← "aaaaaa"
Data[2] ← "bbbbbb"
Data[3] ← "cccccc"
```

Procedure `Process()` manipulates the values in the array.

The procedure is written in pseudocode as follows:

```
PROCEDURE Process(Format : STRING)
   DECLARE Count, Index, L : INTEGER
   DECLARE Result : STRING
   DECLARE C : CHAR

   Result ← "****"

   FOR Count ← 1 TO LENGTH(Format) STEP 2
      C ← MID(Format, Count, 1)
      L ← STR_TO_NUM(MID(Format, Count + 1, 1))

      Index ← (Count + 1) DIV 2

      CASE OF C
         'X' : Result ← TO_UPPER(Data[Index])
         'Y' : Result ← TO_LOWER(Data[Index])
         'Z' : Result ← "**" & Data[Index]
      ENDCASE

      Data[Index] ← LEFT(Result, L)
   NEXT Count

ENDPROCEDURE
```

(a) Complete the trace table by dry running the procedure when it is called as follows:

```
CALL Process("X3Y2W4")
```

Count	C	L	Index	Result	Data[1]	Data[2]	Data[3]

[6]

(b) The procedure is to be modified. If variable `C` is assigned a value other than `'X'`, `'Y'` or `'Z'`, then procedure `Error()` is called and passed the value of variable `C` as a parameter.

This modification can be implemented by adding a **single line** of pseudocode.

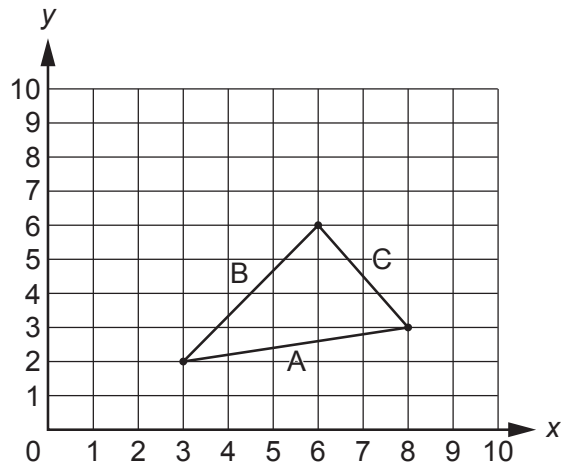
(i) Write the single line of pseudocode.

..... [1]

(ii) State where this new line should be placed.

..... [1]

6 Three points on a grid form a triangle with sides of length A, B and C as shown in the example:



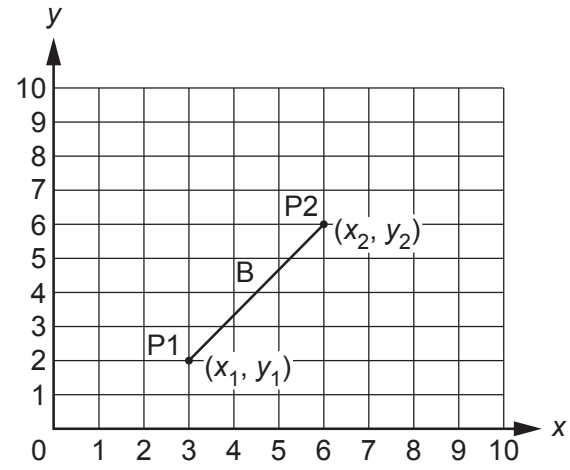
A triangle is said to be right-angled if the following test is true (where A is the length of the longest side):

$A^2 = B^2 + C^2$

A^2 means A multiplied by A, for example 3^2 means 3×3 which evaluates to 9

You can calculate A^2, B^2 and C^2 by using the coordinates of the endpoints of each line.

For example, B^2 is calculated as follows:



The endpoints, P1 and P2, have the coordinates (3, 2) and (6, 6).

The value B^2 is given by the formula:

$B^2 = (x_1 - x_2)^2 + (y_1 - y_2)^2$

In this example:

$B^2 = (3 - 6)^2 + (2 - 6)^2$
$B^2 = (-3)^2 + (-4)^2$
$B^2 = 9 + 16$
$B^2 = 25$

(a) A function `IsRA()` will:

- take three sets of integers as parameters representing the coordinates of the three endpoints that form a triangle
- return `TRUE` if the endpoints form a right-angled triangle, otherwise return `FALSE`.

In pseudocode, the operator '^' represents an exponent, which is the number of times a value is multiplied by itself. For example, the expression $Value^2$ may be written in pseudocode as `Value ^ 2`

Complete the pseudocode for the function `IsRA()`.

```
FUNCTION IsRA(x1, y1, x2, y2, x3, y3 : INTEGER) RETURNS BOOLEAN
```

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```
ENDFUNCTION
```

[6]

(b) The test used to check if a triangle is right-angled can be written in two ways:

$A^2 = B^2 + C^2$

or

$A = \sqrt{(B^2 + C^2)}$

The symbol √ represents the square root operation. For example, √81 = 9

A new function `SQRT()` is written to perform the square root operation. The function takes an integer number as a parameter and returns a positive real value representing the square root of the number.

During testing it is found that the `SQRT()` function returns a value that is only accurate to 4 decimal places.

For example, `SQRT(25)` returns 5.0000125 rather than the correct value of 5.0

The function `IsRA()` from part **(a)** is modified to use the new `SQRT()` function to test if a triangle is right-angled.

Describe a problem that might occur when using the modified `IsRA()` function **and** suggest a solution that still allows the `SQRT()` function to be used.

Problem .....

.....

Solution .....

.....

[2]

7 A fitness club has a computerised membership system. The fitness club offers a number of different exercise classes.

The following information is stored for each club member: name, home address, email address, mobile phone number, date of birth and the exercise(s) they are interested in.

(a) When an exercise class is planned, a new module will send personalised text messages to each member who has expressed an interest in that exercise. Members wishing to join the class send a text message back. Members may decide **not** to receive future text messages by replying with the message 'STOP'.

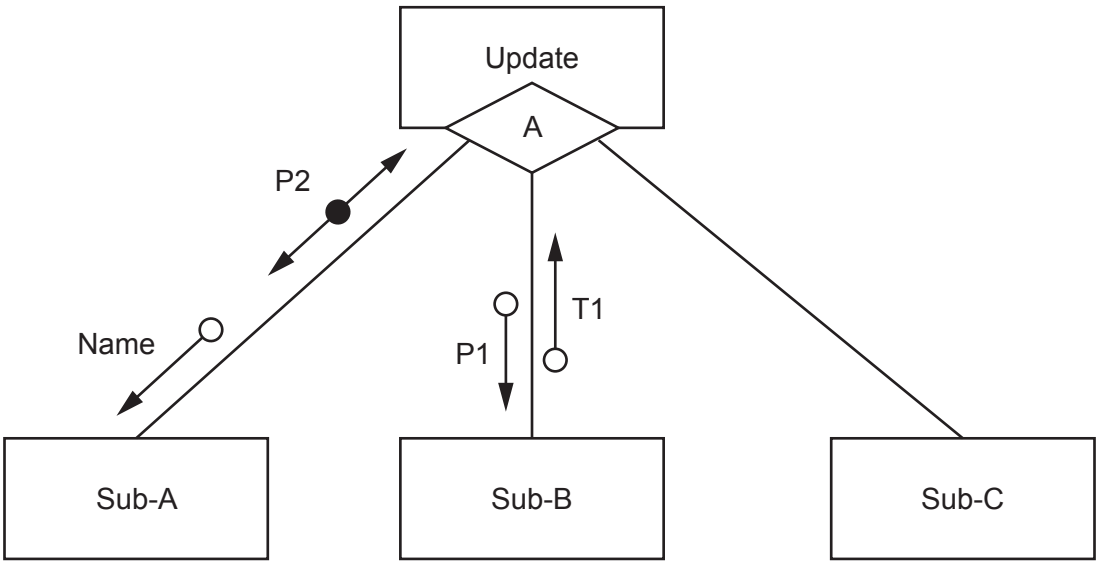
The process of abstraction is used to filter out unnecessary information.

(i) State **one** advantage of applying abstraction to this problem.

.....

..... [1]

(b) The structure chart illustrates part of the membership program:



Data item notes:

- Name contains the name of a club member
- P1 and T1 are of type real.

(i) Explain the meaning of the diamond symbol (labelled with the letter A) in the chart.

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..... [2]

(ii) Write the pseudocode module headers for Sub-A and Sub-B.

Sub-A

.....

.....

Sub-B

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.....

[4]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program is being developed in pseudocode before being converted into a programming language.

(a) The following table shows four valid pseudocode assignment statements.

Complete the table by giving the data type that should be used to declare the variable underlined in each assignment statement.

Assignment statement	Data type
MyVar1 ← Total1 / Total2	
MyVar2 ← 27/10/2023	
MyVar3 ← "Sum1 / Sum2"	
MyVar4 ← Result1 AND Result2	

[4]

(d) The hardware that runs the program is changed and the program needs to be modified so that it works with the new hardware.

Identify the type of maintenance that this represents **and** give **one other** reason why this type of maintenance may be needed.

Type .....

Reason .....

.....

[2]

5 A global 1D array of integers contains four elements, which are assigned values as shown:

```
Mix[1] ← 4
Mix[2] ← 2
Mix[3] ← 3
Mix[4] ← 5
```

A procedure `Process()` manipulates the values in the array.

The procedure is written in pseudocode as follows:

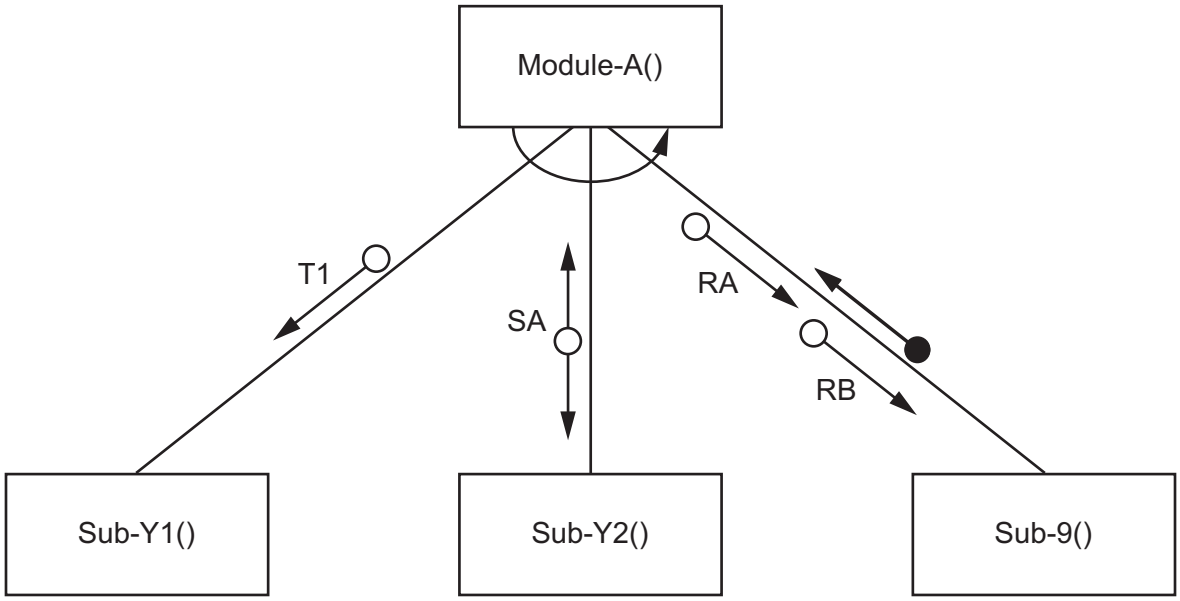
```
PROCEDURE Process(Start : INTEGER)
   DECLARE Value, Index, Total : INTEGER

   Index ← Start
   Total ← 0

   WHILE Total < 20
      Value ← Mix[Index]
      Total ← Total + Value

      IF Index < 4 THEN
         Mix[Index] ← Mix[Index] + Mix[Index+1]
      ELSE
         Mix[Index] ← Mix[Index] + Mix[1]
      ENDIF
```

7 A structure chart shows the modular structure of a program:



(a) Explain the meaning of the curved arrow symbol which begins and ends at Module-A().

.....

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..... [2]

8 A class of students are developing a program to send data between computers. Many computers are connected together to form a wired network. Serial ports are used to connect one computer to another.

Each computer:
- is assigned a unique three-digit ID
- has three ports, each identified by an integer value
- is connected to between one and three other computers.

Messages are sent between computers as a string of characters organised into fields as shown:

```
<STX><DestinationID><SourceID><Data><ETX>
```

Field name	Description
`STX`	a single character marking the start of the message (ASCII value 02)
`DestinationID`	three numeric characters identifying the destination computer
`SourceID`	three numeric characters identifying the source computer
`Data`	a variable length string containing the data being sent (Minimum length is 1 character)
`ETX`	a single character marking the end of the message (ASCII value 03)

For example, the following message contains the data "Hello Jack" being sent from computer "202" to computer "454":

```
<STX>"454202Hello Jack"<ETX>
```

Each computer will run a copy of the same program. Each program will contain a global variable `MyID` of type string which contains the unique ID of the computer in which the program is running.

The first two program modules are defined as follows:

Module	Description
`GetData()` (already written)	• returns the data field from a message that has been received • If no message is available, the module waits until one has been received.
`ReceiveFile()`	• takes a file name as a parameter of type string • creates a text file with the given file name (no checking required) • writes the data field returned by `GetData()` to the file • repeats until the data field is "****", which is not written to the file • outputs a final message giving the total number of characters written to the file, for example: `132456 characters were written to newfile.txt`

(a) Write pseudocode for module `ReceiveFile()`.

Module `GetData()` has already been written and must be used.

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..... [7]

(b) The use of the string "****" as explained in the module description for `ReceiveFile()` may cause a problem.

Explain the problem and suggest a solution.

Problem .....

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Solution .....

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[3]

(c) Two new modules are defined, which will allow two users to exchange messages.

Module	Description
`Transmit()` (already written)	• takes two parameters: ○ a string representing a message ○ an integer representing a port number • transmits the message using the given port
`Chat()`	• takes two parameters: ○ a string representing a Destination ID ○ an integer representing a port number • extracts data from a received message using `GetData()` and outputs it • forms a message using data input by the user and sends it using `Transmit()` • repeats until either the output string or the sent string is `"Bye"`

Reminders:

- Each program contains a global variable `MyID` of type string which contains the unique ID of the computer in which the program is running.
- Messages are sent between computers as a string of characters organised into fields as shown:

```
<STX><DestinationID><SourceID><Data><ETX>
```

Write pseudocode for module `Chat()`.

Modules `GetData()` and `Transmit()` must be used.

[7]

(d) Module `GetData()` returns the data field from a message that has been received. If no message is available, the module waits until one has been received.

Explain the limitation of this on module `Chat()` from part **(c)**.

Describe a modification to `GetData()` to address this limitation.

Limitation .....

.....

.....

.....

Modification .....

.....

.....

.....

[3]

```
        Index ← (Value MOD 4) + 1
    ENDWHILE
    Mix[1] ← Total * Index
ENDPROCEDURE
```

Complete the trace table on the opposite page by dry running the procedure when it is called as follows:

```
CALL Process(2)
```

Index	Value	Total	Mix[1]	Mix[2]	Mix[3]	Mix[4]

[6]

4 A procedure `Count()` will:

1. input a value (all values will be positive integers)
2. count the number of odd values and count the number of even values
3. repeat from step 1 until the value input is 99
4. output the two count values, with a suitable message.

The value 99 must not be counted.

(a) Write pseudocode for the procedure `Count()`.

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..... [6]

(b) The procedure `Count()` is to be tested.

Typical test data would consist of odd and even values, for example:

23, 5, 64, 100, 2002, 1, 8, 900, 99

The purpose of this test would be to test a typical mix of even and odd values and check the totals.

Give **three** test data sequences that may be used to test **different** aspects of the procedure.

Do **not** include invalid data.

Sequence 1:

Test data .....

Purpose of test. .....

.....

Sequence 2:

Test data .....

Purpose of test. .....

.....

Sequence 3:

Test data .....

Purpose of test. .....

.....

[3]

5 A global 1D array of integers contains four elements, which are assigned values as shown:

```
Mix[1] ← 1
Mix[2] ← 3
Mix[3] ← 4
Mix[4] ← 2
```

A procedure `Process()` manipulates the values in the array.

The procedure is written in pseudocode:

```
PROCEDURE Process(Start : INTEGER)
   DECLARE Value, Index, Count : INTEGER

   Index ← Start
   Count ← 0

   REPEAT
      Value ← Mix[Index]
      Mix[Index] ← Mix[Index] - 1
      Index ← Value
      Count ← Count + 1
   UNTIL Count = 5

   Mix[4] ← Count * Index

ENDPROCEDURE
```

Complete the trace table on the opposite page by dry running the procedure when it is called as follows:

```
CALL Process(3)
```

Index	Value	Count	Mix[1]	Mix[2]	Mix[3]	Mix[4]

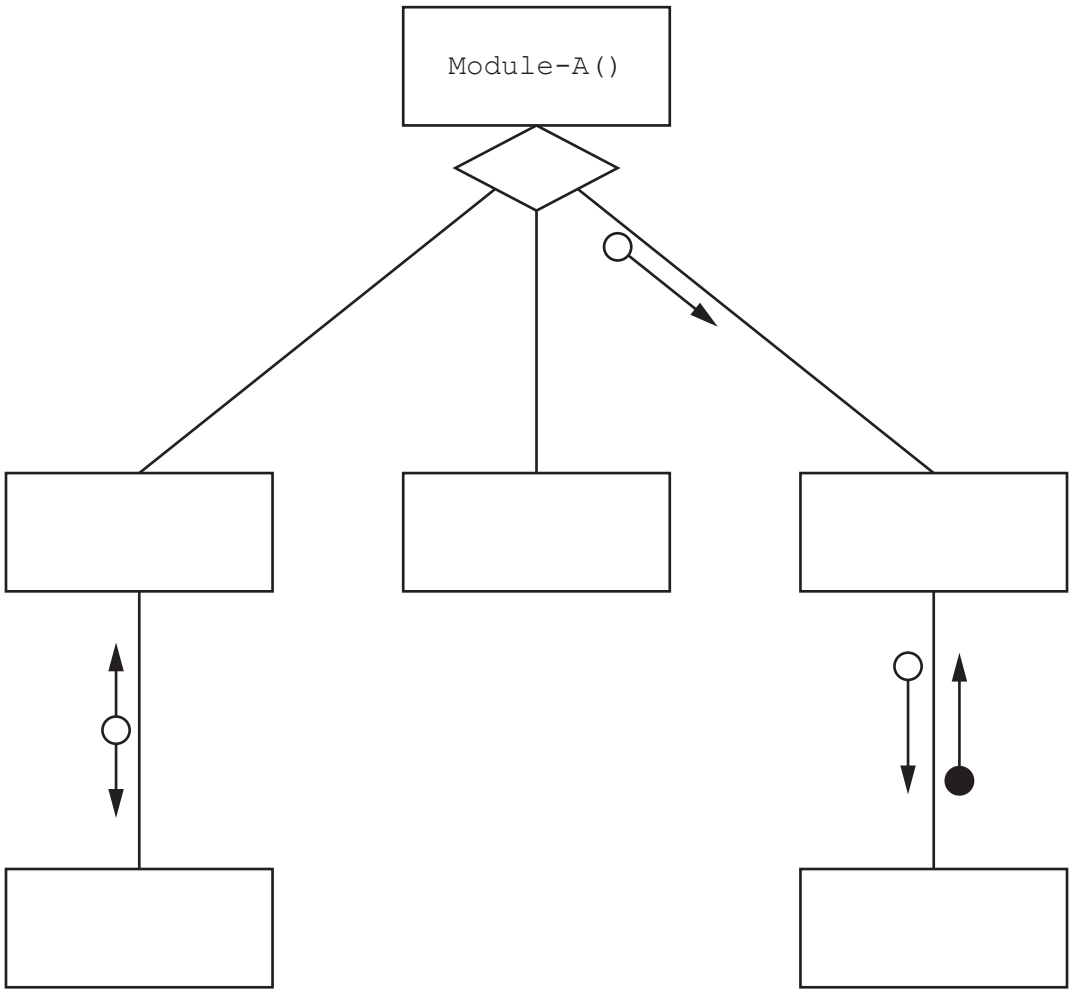
[6]

7 A program contains six modules:

Pseudocode module header
`PROCEDURE Module-A()`
`PROCEDURE Module-X(T1 : INTEGER, S2 : REAL)`
`PROCEDURE Reset(BYREF Code : INTEGER)`
`FUNCTION Restore(OldCode : INTEGER) RETURNS BOOLEAN`
`FUNCTION Module-Y(RA : INTEGER, RB : BOOLEAN) RETURNS BOOLEAN`
`FUNCTION Module-Z(SA : INTEGER) RETURNS INTEGER`

`Module-X()` calls `Reset()`
`Module-Y()` calls `Restore()`

(a) Complete the structure chart for these modules.



[4]

(b) Explain the meaning of the diamond symbol as used in the diagram in part **(a)**.

.....

..... [2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 The following pseudocode represents part of the algorithm for a program:

```
CASE OF ThisValue
   < 30       : Level ← "Low"           // less than 30
                Check ← 1
   < 20       : Level ← "Very Low"      // less than 20
                Check ← ThisValue / 2
   30 TO 40   : Level ← "Medium"        // between 30 and 40
                Check ← ThisValue / 3
                Data[ThisValue] ← Data[ThisValue] + 1
   > 40       : Level ← "High"
ENDCASE
```

(a) Complete the table by writing the answer for each row:

	Answer
The value assigned to `Level` when `ThisValue` is 40	
The value assigned to `Check` when `ThisValue` is 36	
The value assigned to `Level` when `ThisValue` is 18	
The number of elements in array `Data` that may be incremented	

[4]

(b) The pseudocode contains four assignments to variable `Level`. One of these assignments will never be performed.

Identify this assignment **and** explain why this is the case.

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..... [3]

(c) The following line is added immediately before the `ENDCASE` statement:

```
OTHERWISE : Level ← "Undefined"
```

State why this assignment is never performed.

.....

..... [1]

5 An algorithm is designed to find the smallest numeric value from an input sequence and count how many numeric values have been input.
An example of an input sequence is:

23, AB56, 17, 23ZW, 4, 10, END

Numeric input values are all integers and non-numeric input is ignored, except for the string "END" which is used to terminate the sequence.

The algorithm is expressed in pseudocode as shown:

```
DECLARE NextInput : STRING
DECLARE Min, Count, Num : INTEGER

Min ← 999
Count ← 0

REPEAT
   INPUT NextInput
   IF IS_NUM(NextInput) = TRUE THEN
      Num ← STR_TO_NUM(NextInput)
      IF Num > Min THEN
         Min ← Num
      ENDIF
      Count ← Count & 1
   ENDIF
UNTIL NextInput ← "END"

OUTPUT "The minimum value is ", Min, " and the count was ", Count
```

(a) The pseudocode contains three errors due to the incorrect use of **operators**.

Identify each error **and** state the correction required.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(b) The operator errors are corrected and the algorithm is tested as follows:

The input sequence:

```
18, 4, ONE, 27, 189, ERIC, 3, 65, END
```

produces the output:

```
The minimum value is 3 and the count was 6
```

The algorithm is tested with a different test data sequence. The sequence contains a mix of integer and non-numeric values. It is terminated correctly but the algorithm produces unexpected results.

(i) Explain the problem with the algorithm.

.....

.....

.....

..... [2]

(ii) Give a sequence of **four** test data values that could be input to demonstrate the problem.

Value 1 .....

Value 2 .....

Value 3 .....

Value 4 .....

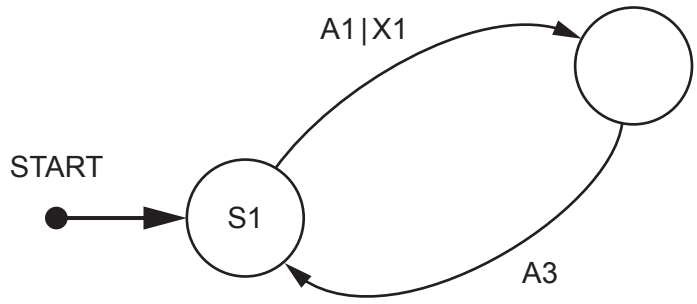
[2]

7 An algorithm is represented by a state-transition diagram.

The table shows the inputs, outputs and states for the algorithm:

Current state	Input	Output	Next state
S1	A1	X1	S2
S2	A3	none	S1
S2	A2	X4	S5
S5	A1	X1	S5
S5	A4	X2	S2
S5	A3	none	S3
S1	A9	X9	S3
S3	A9	X9	S4

Complete the state-transition diagram to represent the information given in the table.



[5]

Refer to the **insert** for the list of pseudocode functions and operators.

1 The following pseudocode represents part of the algorithm for a program.

Line numbers are for reference only.

```
10   DECLARE Sheet4 : ARRAY[1:2, 1:50] OF INTEGER
…
100  FOR PCount ← 0 TO 49
101     Sheet4[1, PCount] ← 0
102     Sheet4[2, PCount] ← 47
103  NEXT PCount
```

(a) The pseudocode contains references to an array.

Complete the table by writing the answer for each row.

	Answer
The dimension of the array	
The name of the variable used as an array index	
The number of elements in the array	

[3]

(b) The pseudocode contains two errors. One error is that variable `PCount` has not been declared.

Identify the **other** error **and** state the line number where it occurs.

Error .....

.....

.....

Line number .....

[2]

5 A program is designed, coded and compiled without errors. The compiled code is sent for testing.

(a) The program will be tested using the walkthrough method.

Additional information will be needed before this method can be used.

Identify this additional information **and** explain why it is needed.

Additional information .....

.....

Explanation .....

.....

.....

.....

[3]

(b) Testing is completed and the program is made available to users.

Some time later, changes are made to the program to improve the speed of response.

State the type of maintenance that has been applied to the program.

..... [1]

7 A school has a library system which allows students to borrow books for a length of time. Information relating to students and books is stored in text files. Student information includes name, home address, email address, date of birth, tutor and subject choices. Book information includes author, title, subject category, library location and the date that the book was borrowed.

A program helps the staff to manage the borrowing of books.

(a) A new module needs to be written to generate emails to send to students who have an overdue book. Students who are sent an email are prevented from borrowing any more books until the overdue book is returned.

The process of abstraction has been used when designing the new module.

(i) State the purpose of applying abstraction to this problem.

.....

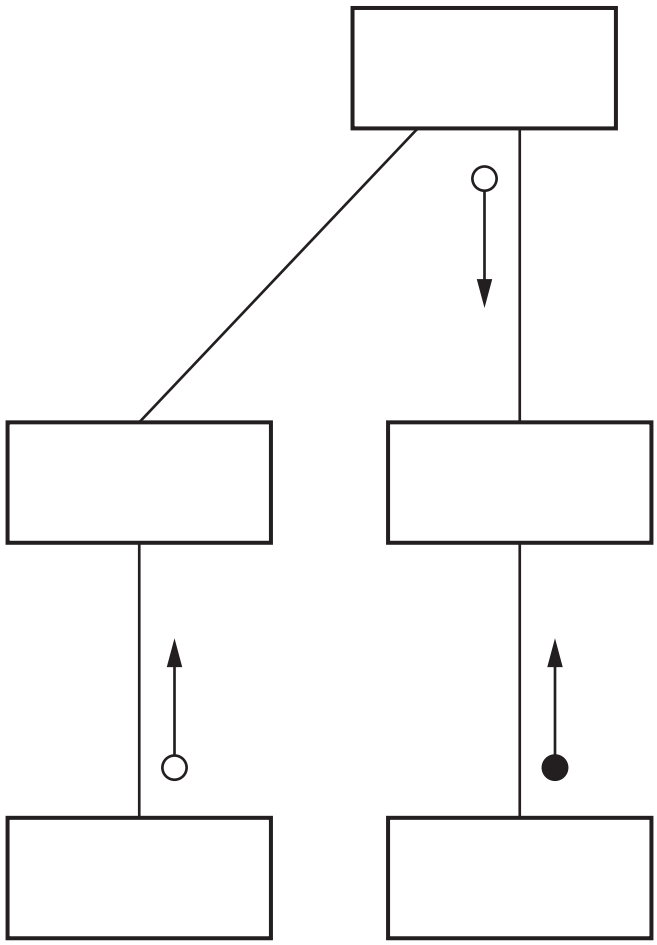
..... [1]

(b) Part of the library program contains program modules with headers as follows:

Pseudocode module header
`PROCEDURE Module-X()`
`PROCEDURE Module-Y(BYREF RA : INTEGER, SA : REAL)`
`PROCEDURE Overlay()`
`FUNCTION Replace(RA : INTEGER, RB : BOOLEAN) RETURNS BOOLEAN`
`FUNCTION Reset(TA : STRING) RETURNS INTEGER`

`Module-X()` and `Module-Y()` are both called from module `Overlay()`.

Complete the structure chart.



[3]

8 A computer shop assembles desktop computers, using items bought from several suppliers. A text file `Stock.txt` contains information about each item.

Information for each item is stored as a single line in the `Stock.txt` file in the format:

```
<ItemNum><SupplierCode><Description>
```

Item information is as follows:

	Format	Comment
`ItemNum`	4 numeric characters	unique number for each item in the range “0001” to “5999” inclusive
`SupplierCode`	3 alphabetic characters	code to identify the supplier of the item
`Description`	a string	a minimum of 12 characters

The file is organised in ascending order of `ItemNum` and does not contain all possible values in the range.

The programmer has defined the first program module as follows:

Module	Description
`ChangeSupp()`	• called with two parameters `Code1` and `Code2` of type string that represent valid supplier codes • creates a new file `NewStock.txt` from the contents of the file `Stock.txt` where any reference to `Code1` is replaced by `Code2` • returns a count of the number of items that have had their supplier code changed

(a) Write pseudocode for module `ChangeSupp()`.

[8]

(b) A new module is required:

Module	Description

`Report_1()`	• takes a parameter of type string that represents a `SupplierCode`
	• searches the `Stock.txt` file for each line of item information that contains the given `SupplierCode`
	• produces a formatted report of items for the given `SupplierCode`, for example, for supplier DRG, the output could be:
	
	`Report for Supplier: DRG`
	
	`Item  Description`
	
	`1234  USB Printer Cable 3m`
	`1273  32GB USB Flash Drive`
	`1350  Mouse Mat 320 x 240mm`
	
	`Number of items listed: 3`

Write pseudocode for module `Report_1()`.

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..... [6]

(c) The format of the output from module `Report_1()` from **part (b)** is changed. The number of items listed is moved to the top of the report as shown in the example:

```
Report for Supplier: DRG
Number of items listed: 3

Item        Description

1234        USB Printer Cable 3m
1273        32GB USB Flash Drive
1350        Mouse Mat 320 x 240mm
```

(i) Explain why this new layout would increase the complexity of the algorithm.

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..... [2]

(ii) The algorithm will be modified to produce the report in the new format. The modified algorithm will be implemented so that the file `Stock.txt` is only read once.

Describe the modified algorithm.

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..... [3]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program calculates the postal cost based on the weight of the item and its destination. Calculations occur at various points in the program and these result in the choice of several possible postal costs. The programmer has built these postal costs into the program.

For example, the postal cost of $3.75 is used in the following lines of pseudocode:

```
IF Weight < 250 AND ValidAddress = TRUE THEN
   ItemPostalCost ← 3.75    // set postal cost for item to $3.75
   ItemStatus ← "Valid"     // item can be sent
ENDIF
```

(a) (i) Identify a more appropriate way of representing the postal costs.

..... [1]

(ii) Describe the advantages of your answer to **part (a)(i)** with reference to this program.

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..... [3]

(b) The lines of pseudocode contain features that make them easier to understand.

State **three** of these features.

1 .....

2 .....

3 .....

[3]

5 A programmer has produced the following pseudocode to output the square root of the numbers from 1 to 10.

Line numbers are for reference only.

```
10  DECLARE Num : REAL
11  Num ← 1.0
...
40  REPEAT
41     CALL DisplaySqrt(Num)
42     Num ← Num + 1.0
43  UNTIL Num > 10
...
50  PROCEDURE DisplaySqrt(BYREF ThisNum : REAL)
51     OUTPUT ThisNum
52     ThisNum ← SQRT(ThisNum)       // SQRT returns the square root
53     OUTPUT " has a square root of ", ThisNum
54  ENDPROCEDURE
```

The pseudocode is correctly converted into program code.

Function `SQRT()` is a library function and contains no errors.

The program code compiles without errors, but the program gives unexpected results. These are caused by a design error.

(a) Explain why the program gives unexpected results.

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..... [3]

(b) Explain why the compiler does **not** identify this error.

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..... [1]

(c) Describe how a typical Integrated Development Environment (IDE) could be used to identify this error.

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..... [3]

(d) The pseudocode is converted into program code as part of a larger program.

During compilation, a complex statement generates an error.

The programmer does not want to delete the complex statement but wants to change the statement so that it is ignored by the compiler.

State how this may be achieved.

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..... [1]

7 A computer system for a shop stores information about each customer. The items of information include name and address (both postal and email) together with payment details and order history. The system also stores the product categories they are interested in and how they would like to be contacted.

(a) The shop wants to add a program module that will generate emails to be sent to customers who may be interested in receiving details of new products.

(i) State **three** items of information that the new module would need. Justify your choice in each case.

Information .....

Justification .....

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Information .....

Justification .....

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Information .....

Justification .....

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[3]

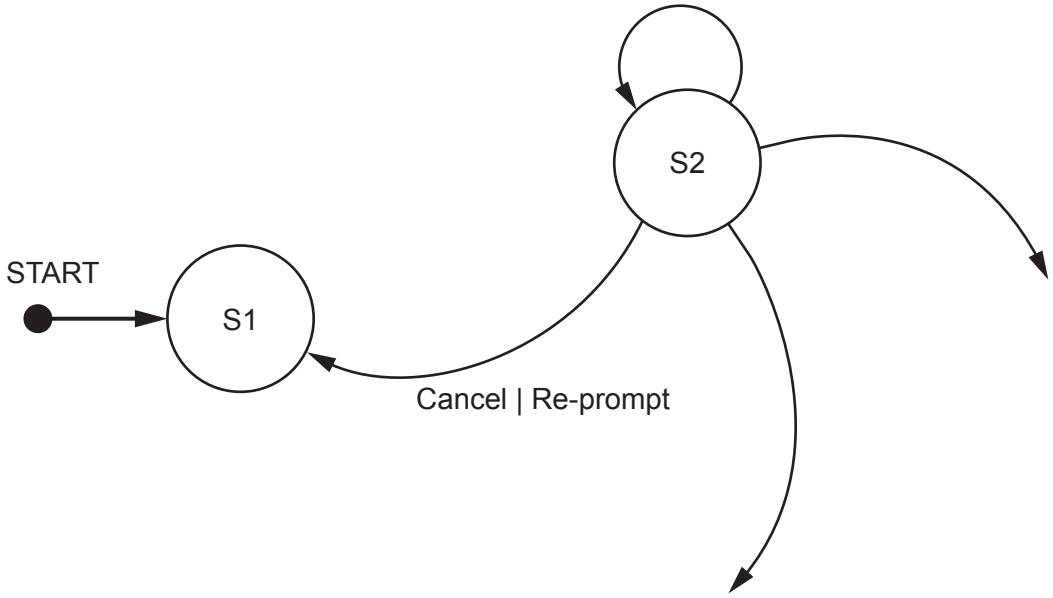
(b) The program includes a module to validate a Personal Identification Number (PIN). This is used when customers pay for goods using a bank card.

A state-transition diagram has been produced for this module.

The table show the inputs, outputs and states for this part of the program:

Current state	Input	Output	Next state
S1	Input PIN		S2
S2	Re-input PIN	Display error	S2
S2	Cancel	Re-prompt	S1
S2	Valid PIN	Enable payment	S4
S2	Too many tries	Block Account	S3

Complete the state-transition diagram to represent the information given in the table.



[4]

8 A computer shop assembles computers using items bought from several suppliers. A text file `Stock.txt` contains information about each item.

Information for each item is stored as a single line in the `Stock.txt` file in the format:

```
<ItemNum><SupplierCode><Description>
```

Valid item information is as follows:

	Format	Comment
`ItemNum`	4 numeric characters	unique number for each item in the range ″0001″ to ″5999″ inclusive
`SupplierCode`	3 alphabetic characters	to identify the supplier of the item
`Description`	a string	a minimum of 12 characters

The file is organised in ascending order of `ItemNum` and does **not** contain all possible values in the range.

A programmer has started to define program modules as follows:

Module	Description
`OnlyAlpha()` (already written)	• called with a parameter of type string • returns `TRUE` if the string contains only alphabetic characters, otherwise returns `FALSE`
`CheckInfo()`	• called with a parameter of type string representing a line of item information • checks to see whether the item information in the string is valid • returns `TRUE` if the item information is valid, otherwise returns `FALSE`

(a) Write pseudocode for module `CheckInfo()`.

Module `OnlyAlpha()` should be used as part of your solution.

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.....

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..... [7]

(c) The program contains modules `SuppExists()` and `CheckSupplier()`. These have been written but contain errors. These modules are called from several places in the main program and testing of the main program (integration testing) has had to stop.

Identify a method that can be used to continue testing the main program before the errors in these modules have been corrected **and** describe how this would work.

Method .....

Description .....

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.....

.....
[3]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A programmer has written a program which includes the function `Calculate()`.
When the program is run, the function returns an unexpected value.

(a) Describe how a typical Integrated Development Environment (IDE) could be used to help debug the program to find the errors in the function `Calculate()`.

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..... [4]

(b) The algorithm for function `Calculate()` contains the three pseudocode statements shown.

Describe the error in each statement or write 'no error' if the statement contains no error.

Assume any variables used are of the correct type for the given function.

Statement 1: `Index ← STR_TO_NUM(("27") + 2)`

Error .....

.....

Statement 2: `Index ← STR_TO_NUM(MID("CPE1704TKS", 4, 2))`

Error .....

.....

Statement 3: `IF MONTH(ThisDate) > '6' THEN`

Error .....

.....
[3]

5 Several companies are developing websites to market a new type of games console. The company that is first to create a website that can demonstrate the interactive features of the games console will have an advantage over the others. The requirements for the website are likely to change as more information about the features of the console are made available.

One company has decided to develop their website using a program development life cycle based on the waterfall model.

(a) (i) Give **two** reasons why this may **not** be the most appropriate model to use in this case.

Reason 1 .....

.....

Reason 2 .....

.....
[2]

(ii) Identify a **more appropriate** program development life cycle model for this scenario.

..... [1]

(b) The website has been running in test mode for several weeks.

Identify **and** describe a final stage of testing that should take place before the website is made available to all customers.

Stage .....

Description .....

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.....
[3]

7 A school has a computerised library system that allows students to borrow books for a fixed length of time. The system uses text files to store details of students, books and loans.

A new module is to be written which will generate emails to each student who has an overdue book.

(a) Decomposition will be used to break down the problem of designing the new module into sub-problems.

Identify **three** program modules that could be used in the design **and** describe their use.

Module 1 .....

Use .....

.....

.....

.....

Module 2 .....

Use .....

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.....

.....

Module 3 .....

Use .....

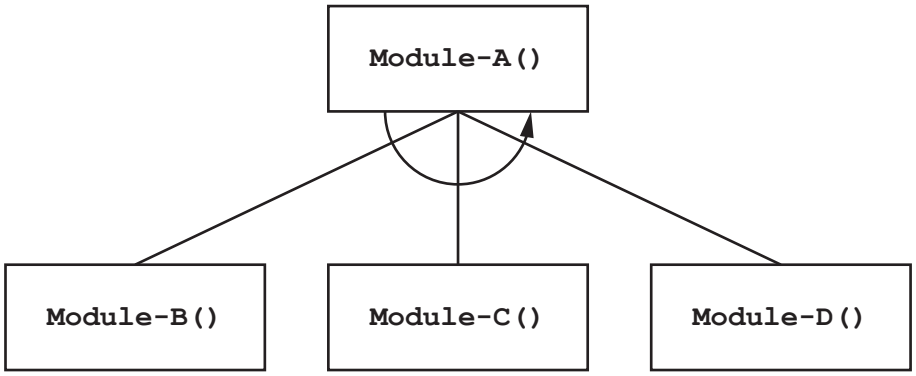
.....

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[3]

(b) The program designer produces a structure chart for the new module. Part of the structure chart is shown:



(i) Explain the relationship between the four modules shown.

.....

.....

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..... [2]

(ii) Two new modules are added: `Module-X()` and `Module-Y()`.

- `Module-X()` has no parameters.
- `Module-Y()` will take a string and a real number as parameters and return a Boolean value.
- `Module-D()` will call either `Module-X()` or `Module-Y()`.

Draw **only** the part of the structure chart that represents the relationship between `Module-X()`, `Module-Y()` and `Module-D()`.

[3]

8 A computer shop assembles computers using items bought from several suppliers. A text file `Stock.txt` contains information about each item.

Information for each item is stored as a single line in the `Stock.txt` file in the format:

```
<ItemNum><SupplierCode><Description>
```

Item information is as follows:

	Format	Comment
`ItemNum`	4 numeric characters	unique for each item in the range "0001" to "5999" inclusive
`SupplierCode`	5 alphabetic characters	to identify the supplier of the item
`Description`	a string	a minimum of 12 characters

The file is organised in ascending order of `ItemNum` and does **not** contain all possible values in the range.

A programmer has started to define program modules as follows:

Module	Description
`SuppExists()` (already written)	• called with a parameter of type string representing a supplier code • returns `TRUE` if the supplier code is already in use, otherwise returns `FALSE`
`IsNewSupp()`	• called with a parameter of type string representing a new supplier code • returns `TRUE` if the string only contains alphabetic characters (either upper or lower case) and the supplier code is **not** already in use, otherwise returns `FALSE`

(a) Write pseudocode for module `IsNewSupp()`.

Module `SuppExists()` has already been written and should be used as part of your solution. Module `SuppExists()` will generate a run-time error if the given parameter is not 5 characters in length.

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..... [7]

(b) A new module has been defined:

Module	Description
`CheckNewItem()`	• called with a parameter of type string representing a line of item information • checks to see whether an item with the same `ItemNum` already exists in the file • returns `TRUE` if the `ItemNum` is not already in the file, otherwise returns `FALSE`

Write **efficient** pseudocode for module `CheckNewItem()`.

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.....

.....

..... [7]

(c) The program modules `SuppExists()`, `IsNewSupp()` and `CheckNewItem()` are part of a group of modules that are combined to create a complete stock control program.

Each module in the program is tested individually during development and is debugged as necessary. It is then added to the program and further testing performed.

(i) Identify this method of testing.

..... [1]

(ii) One of the modules does not work properly when it is added to the program.

Describe a testing method that can be used to address this problem so that testing can continue and other modules can be added.

.....

.....

.....

..... [2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 A program is required for a shopping website.

(a) Part of the program requires four variables. The following table describes the use of each variable.

Complete the table by adding the most appropriate data type for each variable.

Variable use	Data type
Store the number of days in the current month	
Store the first letter of the customer's first name	
Store an indication of whether a year is a leap year	
Store the average amount spent per customer visit	

[4]

(b) The designer considers the use of a development life cycle to split the development of the website into several stages.

(i) State **one** benefit of a development life cycle when developing the website.

.....

..... [1]

(ii) Analysis is one stage of a development life cycle.

State **one** document that may be produced from the analysis stage of the website project.

.....

..... [1]

(c) The program will be developed using the Rapid Application Development (RAD) life cycle.

(i) State **one** principle of this life cycle.

.....

..... [1]

(ii) Give **two** benefits and **one** drawback of its use compared to the waterfall life cycle.

Benefit 1 .....

.....

Benefit 2 .....

.....

Drawback .....

.....

[3]

(d) Adaptive maintenance needs to be carried out on the website program.

Give **two** reasons why adaptive maintenance may be required.

1 .....

.....

2 .....

.....

[2]

5 **(a)** A text string contains three data items concatenated as shown:

<StockID><Description><Cost>

Item lengths are:

Item	Length
StockID	5
Description	32
Cost	the remainder of the string

A procedure `Unpack()` takes four parameters of type string. One parameter is the original text string. The other three parameters are used to represent the three data items shown in the table and are assigned values within the procedure. These values will be used by the calling program after the procedure ends.

(i) Write pseudocode for the procedure `Unpack()`.

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..... [6]

(c) `Unpack()` is part of a program made up of several modules. During the design stage, it is important to follow good programming practice. One example of good practice is the use of meaningful identifier names.

Give the reason why this is good practice. Give **two other** examples of good practice.

Reason .....

.....

.....

Example 1 .....

.....

.....

Example 2 .....

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.....
[3]

(d) The program that includes `Unpack()` is tested using the walkthrough method.

Describe this method **and** explain how it can be used to identify an error.

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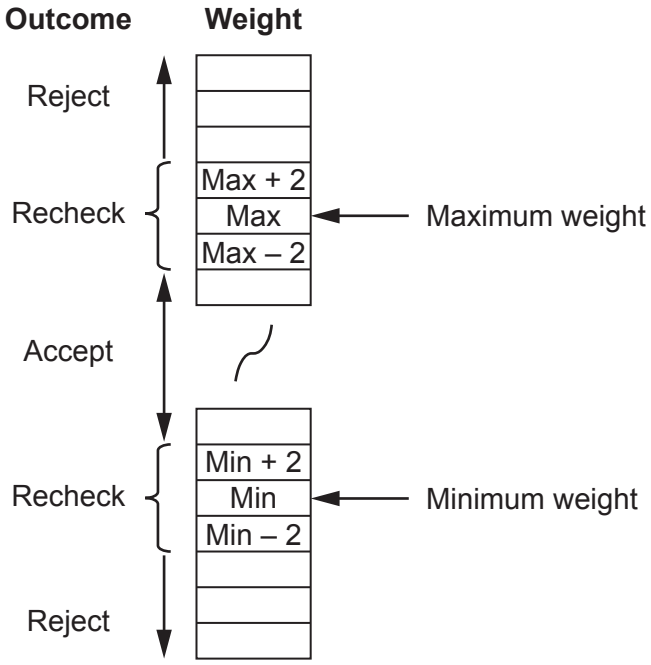
..... [3]

6 Components are weighed during manufacture. Weights are measured to the nearest whole gram.

Components that weigh at least 3 grams more than the maximum weight, or at least 3 grams less than the minimum weight, are rejected.

A component is rechecked if it weighs within 2 grams of either the maximum or minimum weight.

The final outcome of weighing each component is shown below:



A function `Status()` will be called with three parameters. These are integers representing the weight of an individual component together with the minimum and maximum weights.

The value returned from the function will be as follows:

Outcome	Return value
Accept	`'A'`
Reject	`'R'`
Recheck	`'C'`

(a) Complete the following test plan for **five** tests that could be performed on function `Status()`. The tests should address all possible outcomes.

Test number	Component weight	Min	Max	Expected return value
1				`'A'`
2				
3				
4				
5				

[5]

Refer to the **insert** for the list of pseudocode functions and operators.

1 **(a)** A programmer is developing an algorithm to solve a problem. Part of the algorithm would be appropriate to implement as a subroutine (a procedure or a function).

(i) State **two** reasons why the programmer may decide to use a subroutine.

1 .....

.....

2 .....

.....

[2]

(b) The algorithm in **part (a)** is part of a program that will be sold to the public.
All the software errors that were identified during in-house testing have been corrected.

Identify **and** describe the additional test stage that may be carried out before the program is sold to the public.

Test stage .....

Description .....

.....

.....

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.....

.....

[4]

2 **(a)** An algorithm will process data from a test taken by a group of students. The algorithm will prompt and input the name and test mark for each of the 35 students.

The algorithm will add the names of all the students with a test mark of less than 20 to an existing text file `Support_List.txt`, which already contains data from other group tests.

(i) Describe the steps that the algorithm should perform.

Do **not** include pseudocode statements in your answer.

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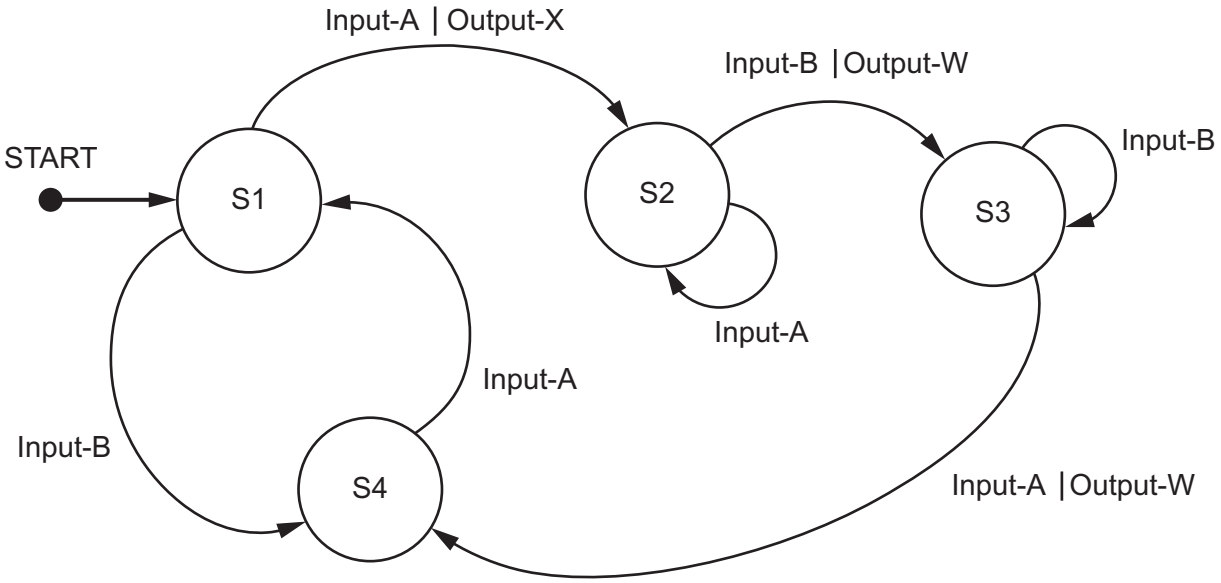
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..... [5]

(b) Examine the following state-transition diagram.



Complete the table to show the inputs, outputs and next states.

Input	Output	Next state
		S1
Input-A		
		S2
	Output-W	
	Output-W	

[4]

6 **(a)** The factorial of an integer number is the product of all the integers from that number down to 1.

In general, the factorial of n is n × (n−1) × ... × 2 × 1

For example, the factorial of 5 is 5 × 4 × 3 × 2 × 1 = 120

In this question, n will be referred to as the `BaseNumber`.

A function `FindBaseNumber()` will:

- be called with a positive, non-zero integer value as a parameter
- return `BaseNumber` if the parameter value is the factorial of the `BaseNumber`
- return −1 if the parameter value **is not** a factorial.

For example:

Parameter value	Value returned
120	5
12	−1
6	3
1	1

`FindBaseNumber(12)` will return −1 because 12 is not a factorial.

You may use the rest of this page for rough working.

Write pseudocode for the function `FindBaseNumber()`.

[7]

(b) A program is written to allow a user to input a sequence of values to be checked using the function `FindBaseNumber()`.

The user will input one value at a time. The variable used to store the user input has to be of type string because the user will input 'End' to end the program.

Valid input will be converted to an integer and passed to `FindBaseNumber()` and the return value will be output.

Complete the table by giving **four** invalid strings that may be used to test distinct aspects of the required validation. Give the reason for your choice in each case.

Input	Reason for choice
	....
	....
	....
	....

[4]

Refer to the **insert** for the list of pseudocode functions and operators.

1 **(a)** An algorithm includes a number of complex calculations. A programmer is writing a program to implement the algorithm and decides to use library routines to provide part of the solution.

State **three** possible benefits of using library routines in the development of the program.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) The pseudocode statements in the following table may contain errors.

State the error in each case or write 'NO ERROR' if the statement contains no error.

Assume that any variables used are of the correct type for the given function.

Statement	Error
`IF EMPTY ← "" THEN`	
`Status ← IS_NUM(-23.4)`	
`X ← STR_TO_NUM("37") + 5`	
`Y ← STR_TO_NUM("37" + "5")`	

[4]

6 The following pseudocode algorithm attempts to check whether a string is a valid email address.

```
FUNCTION IsValid(InString : STRING) RETURNS BOOLEAN
   DECLARE Index, Dots, Ats, Others : INTEGER
   DECLARE NextChar : CHAR
   DECLARE Valid : BOOLEAN

   Index ← 1
   Dots ← 0
   Ats ← 0
   Others ← 0
   Valid ← TRUE

   REPEAT
      NextChar ← MID(InString, Index, 1)
      CASE OF NextChar
         '.' : Dots ← Dots + 1
         '@' : Ats ← Ats + 1
               IF Ats > 1 THEN
                  Valid ← FALSE
               ENDIF
         OTHERWISE : Others ← Others + 1
      ENDCASE

      IF Dots > 1 AND Ats = 0 THEN
         Valid ← FALSE
      ELSE
         Index ← Index + 1
      ENDIF

   UNTIL Index > LENGTH(InString) OR Valid = FALSE

   IF NOT (Dots >= 1 AND Ats = 1 AND Others > 8) THEN
      Valid ← FALSE
   ENDIF

   RETURN Valid

ENDFUNCTION
```

(a) Part of the validation is implemented by the line:

```
IF NOT (Dots >= 1 AND Ats = 1 AND Others > 8) THEN
```

State the values that would result in the condition evaluating to `TRUE`.

.....

.....

..... [1]

(b) (i) Complete the trace table by dry running the function when it is called as follows:

```
Result ← IsValid("Liz.123@big@net")
```

Index	NextChar	Dots	Ats	Others	Valid

[5]

(ii) State the value returned when `IsValid()` is called using the expression shown in **part (b)(i)**.

..... [1]

7 A simple arithmetic expression is stored as a string in the format:

<Value1><Operator><Value2>

An operator character is one of the following: `'+'` `'-'` `'*'` `'/'`

Example arithmetic expression strings:

```
"803+1904"
"34/7"
```

(a) A procedure `Calculate()` will:

- take an arithmetic expression string as a parameter
- evaluate the expression
- output the result.

Assume:

- the string contains only numeric digits and a single operator character
- Value1 and Value2 represent integer values
- Value1 and Value2 are unsigned (they will not be preceded by `'+'` or `'-'`).

(i) Write pseudocode for the procedure `Calculate()`.

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..... [7]

(b) A string representing an arithmetic expression could be in the correct format but be impossible to evaluate.

Give an example of a correctly formatted string and explain why evaluation would be impossible.

Example string .....

Explanation .....

.....

.....

[2]

8 A teacher is designing a program to perform simple syntax checks on programs written by students. Student programs are submitted as text files, which are known as project files.

A project file may contain blank lines.

The teacher has defined the first program module as follows:

Module	Description
`CheckFile()`	• takes the name of an existing project file as a parameter of type string • returns `TRUE` if the file is valid (it contains at least 10 non-blank lines), otherwise returns `FALSE`

(a) Write pseudocode for module `CheckFile()`.

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..... [7]

Further modules are defined as follows:

Module	Description
`CheckLine()`	• takes a line from a project file as a parameter of type string • returns zero if the line is blank or contains no syntax error, otherwise returns an error number as an integer
`CountErrors()`	• takes two parameters: ○ the name of a project file as a string ○ the maximum number of errors as an integer • uses `CheckFile()` to test the project file. Outputs an error message and ends if the project file is not valid • calls `CheckLine()` for each line in the project file • counts the number of errors • outputs the number of errors or a warning message if the maximum number of errors is exceeded

(b) `CountErrors()` is called to check the project file `Jim01Prog.txt` and to stop if more than 20 errors are found.

Write the pseudocode statement for this call.

.....

..... [2]

(d) Module `CheckLine()` includes a check for syntax errors.

Two examples of syntax error that **cannot** be detected from examining a **single** line are those involving selection and iteration.

Give **two other** examples.

1 .....

.....

.....

2 .....

.....

.....

[2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 Sylvia is testing a program that has been written by her colleague. Her colleague tells her that the program does not contain any syntax errors.

(a) (i) State what her colleague means by “does not contain any syntax errors”.

.....

.....

..... [1]

(ii) Identify **and** describe **one** other type of error that the program may contain.

Type of error .....

Description .....

.....

[2]

4 A program controls the heating system in a sports hall.

Part of the program involves reading a value from a sensor. The sensor produces a numeric value that represents the temperature. The value is an integer, which should be in the range 0 to 40 inclusive.

A program function has been written to validate the values from the sensor.

(a) A test plan is needed to test the function.

Complete the table. The first line has been completed for you.

You can assume that the sensor will generate only integer data values.

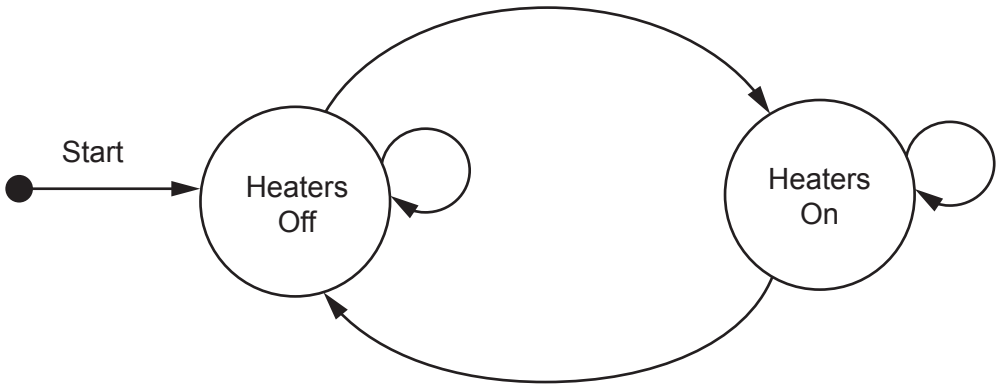
Test	Test data value	Explanation	Expected outcome
1	23	Normal data	Data is accepted
2			
3			
4			
5			

[4]

(b) A program module controls the heaters. This module operates as follows:

- If the temperature is below 10, switch the heaters on.
- If the temperature is above 20, switch the heaters off.

Complete the following state-transition diagram for the heating system:



[3]

5 The following data items will be recorded each time a student successfully logs on to the school network:

Data item	Example data
Student ID	`"CJL404"`
Host ID	`"Lib01"`
Time and date	`"08:30, June 01, 2021"`

The Student ID is six characters long. The other two data items are of variable length.

A single string will be formed by concatenating the three data items. A separator character will need to be inserted between items two and three.

For example:

```
"CJL404Lib01<separator>08:30, June 01, 2021"
```

Each string represents one log entry.

A programmer decides to store the concatenated strings in a 1D array `LogArray` that contains 2000 elements. Unused array elements will contain an empty string.

(a) Suggest a suitable separator character **and** give a reason for your choice.

Character .....

Reason .....

.....

[2]

(b) The choice of data structure was made during one stage of the program development life cycle.

Identify this stage.

..... [1]

2 **(a)** An algorithm will:

1. input an integer value
2. jump to step 6 if the value is zero
3. sum and count the positive values
4. sum and count the negative values
5. repeat from step 1
6. output the two sum values and the two count values.

Draw a program flowchart on the following page to represent the algorithm.

Note that variable declarations are not required in program flowcharts.

[5]

(b) A software company is working on a project to develop a website for a school.

The school principal has some ideas about the appearance of the website but is unclear about all the details of the solution. The principal would like to see an initial version of the website.

(i) Identify a life cycle method that would be appropriate in this case.

Give a reason for your choice.

Life cycle method .....

Reason .....

.....

.....

.....

[2]

(ii) The website project has progressed to the design stage.

State **three** activities that will take place during the design stage of the program development life cycle.

1 .....

2 .....

3 .....

[3]

4 The following is a procedure design in pseudocode.

Line numbers are given for reference only.

```
10  PROCEDURE Check(InString : STRING)
11     DECLARE Odds, Evens, Index : INTEGER
12
13     Odds ← 0
14     Evens ← 0
15     Index ← 1
16
17     WHILE Index <= LENGTH(InString)
18        IF STR_TO_NUM(MID(InString, Index, 1)) MOD 2 <> 0 THEN
19           Odds ← Odds + 1
20        ELSE
21           Evens ← Evens + 1
22        ENDIF
23        Index ← Index + 1
24     ENDWHILE
25
26     CALL Result(Odds, Evens)
27  ENDPROCEDURE
```

(a) Complete the following table by giving the answers, using the given pseudocode.

	Answer
A line number containing a variable being incremented	
The type of loop structure	
The number of functions used	
The number of parameters passed to `STR_TO_NUM()`	
The name of a procedure other than `Check()`	

[5]

(b) The pseudocode includes several features that make it easier to read and understand.

Identify **three** of these features.

1 .....

2 .....

3 .....

[3]

Refer to the **insert** for the list of pseudocode functions and operators.

1 Sylvia is testing a program that has been written by her colleague. Her colleague tells her that the program does not contain any syntax errors.

(a) (i) State what her colleague means by “does not contain any syntax errors”.

.....

.....

..... [1]

(ii) Identify **and** describe **one** other type of error that the program may contain.

Type of error .....

Description .....

.....

[2]

4 A program controls the heating system in a sports hall.

Part of the program involves reading a value from a sensor. The sensor produces a numeric value that represents the temperature. The value is an integer, which should be in the range 0 to 40 inclusive.

A program function has been written to validate the values from the sensor.

(a) A test plan is needed to test the function.

Complete the table. The first line has been completed for you.

You can assume that the sensor will generate only integer data values.

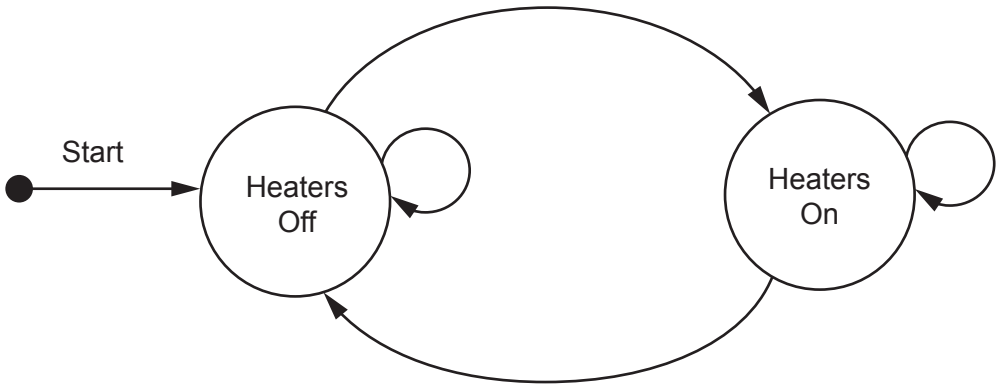
Test	Test data value	Explanation	Expected outcome
1	23	Normal data	Data is accepted
2			
3			
4			
5			

[4]

(b) A program module controls the heaters. This module operates as follows:

- If the temperature is below 10, switch the heaters on.
- If the temperature is above 20, switch the heaters off.

Complete the following state-transition diagram for the heating system:



[3]

5 The following data items will be recorded each time a student successfully logs on to the school network:

Data item	Example data
Student ID	`"CJL404"`
Host ID	`"Lib01"`
Time and date	`"08:30, June 01, 2021"`

The Student ID is six characters long. The other two data items are of variable length.

A single string will be formed by concatenating the three data items. A separator character will need to be inserted between items two and three.

For example:

```
"CJL404Lib01<separator>08:30, June 01, 2021"
```

Each string represents one log entry.

A programmer decides to store the concatenated strings in a 1D array `LogArray` that contains 2000 elements. Unused array elements will contain an empty string.

(a) Suggest a suitable separator character **and** give a reason for your choice.

Character .....

Reason .....

.....

[2]

(b) The choice of data structure was made during one stage of the program development life cycle.

Identify this stage.

..... [1]

Refer to the **insert** for the list of pseudocode functions and operators.

1 (a) A program is being developed to help manage the membership of a football club.

Complete the following identifier table.

Example value	Explanation	Variable name	Data type
`"Wong"`	The preferred name of the member joining the football club		
`FALSE`	A value to indicate whether an existing member of the club lives at the same address		
`19/02/1983`	When the member joined the football club		
`1345`	The number of points a member has earned. Members of the club earn points for different activities.		

[4]

(b) Each pseudocode statement in the following table may contain an error due to the incorrect use of the function or operator.

Describe the error in each case, or write 'NO ERROR' if the statement contains no error.

You can assume that none of the variables referenced are of an incorrect type.

Statement	Error
`Result ← 2 & 4`	
`SubString ← MID("pseudocode", 4, 1)`	
`IF x = 3 OR 4 THEN`	
`Result ← Status AND INT(x/2)`	
`Message ← "Done" + LENGTH(MyString)`	

[5]

2 **(a)** Four program modules form part of a program for a library.

A description of the relationship between the modules is summarised as follows:

Module name	Description
`UpdateLoan()`	• Calls either `LoanExtend()` or `LoanReturn()`
`LoanExtend()`	• Called with parameters `LoanID` and `BookID` • Calls `CheckReserve()` to see whether the book has been reserved for another library user • Returns `TRUE` if the loan has been extended, otherwise returns `FALSE`
`CheckReserve()`	• Called with `BookID` • Returns `TRUE` if the book has been reserved, otherwise returns `FALSE`
`LoanReturn()`	• Called with parameters `LoanID` and `BookID` • Returns a `REAL` (which is the value of the fine to be paid in the case of an overdue loan)

Draw a structure chart to show the relationship between the four modules and the parameters passed between them.

[5]

4 Study the following pseudocode. Line numbers are for reference only.

```
10 FUNCTION Convert(Name : STRING) RETURNS STRING
11
12   DECLARE Flag: BOOLEAN
13   DECLARE Index : INTEGER
14   DECLARE ThisChar : CHAR
15   DECLARE NewName : STRING
16
17   CONSTANT SPACECHAR = ' '
18
19   Flag ← TRUE
20   Index ← 1
21   NewName ← ""      // formatted name string
22
23   WHILE Index <= LENGTH(Name)
24      ThisChar ← MID(Name, Index, 1)
25      IF Flag = TRUE THEN
26         NewName ← NewName & UCASE(ThisChar)
27         IF ThisChar <> SPACECHAR THEN
28            Flag ← FALSE
29         ENDIF
30      ELSE
31         NewName ← NewName & ThisChar
32      ENDIF
33      IF ThisChar = SPACECHAR THEN
34         Flag ← TRUE
35      ENDIF
36      Index ← Index + 1
37   ENDWHILE
38
39   RETURN NewName
40
41 ENDFUNCTION
```

(a) Complete the trace table below by dry running the function when it is called as follows:

```
Result ← Convert("∇in∇a∇∇Cup")
```

Note: The symbol `'∇'` has been used to represent a space character.
Use this symbol for any space characters in the trace table.

The first row has been completed for you.

Name	Flag	Index	NewName	ThisChar
"∇in∇a∇∇Cup"				

[5]

(c) Two changes need to be made to the algorithm.

Change 1: Convert to lower case any character that is not the first character after a space.
Change 2: Replace multiple spaces with a single space.

(i) Change 1 may be implemented by modifying one line of the pseudocode.

Write the modified line.

.....

.....[1]

(ii) Change 2 may be implemented by moving one line of the pseudocode.

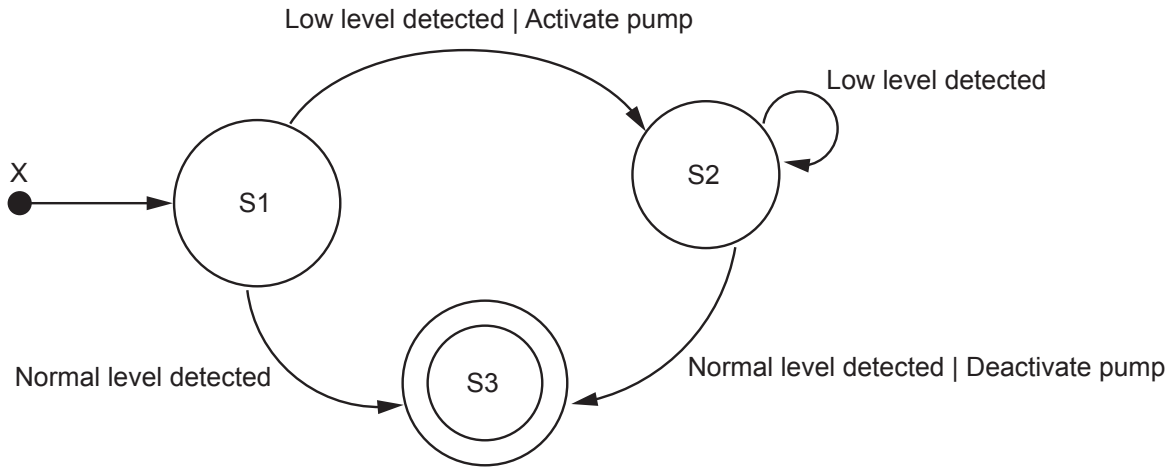
Write the number of the line to be moved and state its new position.

Line number .....

New position .....

.....
[2]

2 **(a)** Examine the following state-transition diagram.



(i) Complete the table with reference to the diagram.

	Answer
The number of transitions that result in a different state	
The number of transitions with associated outputs	
The label that should replace 'X'	
The final or halting state	

[4]

(ii) The current state is S1. The following inputs occur.

1. Low level detected
2. Low level detected
3. Low level detected
4. Low level detected

Give the number of outputs and the current state.

Number of outputs .....

Current state .....

[2]

5 **(a)** A student is learning about arrays.

She wants to write a program to:

- declare a 1D array `RNum` of 100 elements of type `INTEGER`
- assign each element a random value in the range 1 to 200 inclusive
- count and output how many numbers generated were between 66 and 173 inclusive.

(i) Write pseudocode to represent the algorithm.

.....

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.....

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.....

.....

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..... [6]

(ii) The student decides to modify the algorithm so that each element of the array will contain a unique value.

Describe the changes that the student needs to make to the algorithm.

.....

.....

.....

.....

.....

..... [3]

7 A procedure, `FormatName()`:

- is called with a string containing words and spaces as its parameter. In this context, a word is any sequence of characters that does not contain a space character.
- creates a new formatted string from this string with the following requirements:
    1. Any leading spaces removed (spaces before the first word).
    2. Any trailing spaces removed (spaces after the last word).
    3. Any multiple spaces between words converted to a single space.
    4. All characters converted to lower case.

The `FormatName()` procedure has been written in a programming language and is to be tested using the black-box method.

(a) Give a test string that could be used to show that all **four** formatting requirements have been applied correctly.

Use the symbol '∇' to represent a space character.

..... [3]

(b) The `FormatName()` procedure should assign a value to the global variable `FString`.

There is a fault in the program, which means that the assignment does not always take place.

Explain **two** ways of exposing the fault.

.....

.....

.....

.....

..... [2]

Refer to the **insert** for the list of pseudocode functions and operators.

1 (a) A program is being developed to help manage the membership of a football club.

Complete the following identifier table.

Example value	Explanation	Variable name	Data type
`"Wong"`	The preferred name of the member joining the football club		
`FALSE`	A value to indicate whether an existing member of the club lives at the same address		
`19/02/1983`	When the member joined the football club		
`1345`	The number of points a member has earned. Members of the club earn points for different activities.		

[4]

(b) Each pseudocode statement in the following table may contain an error due to the incorrect use of the function or operator.

Describe the error in each case, or write 'NO ERROR' if the statement contains no error.

You can assume that none of the variables referenced are of an incorrect type.

Statement	Error
`Result ← 2 & 4`	
`SubString ← MID("pseudocode", 4, 1)`	
`IF x = 3 OR 4 THEN`	
`Result ← Status AND INT(x/2)`	
`Message ← "Done" + LENGTH(MyString)`	

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```

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Line number .....

New position .....

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[2]